

HAWK with Hint: Algebraic Key Recovery from Side-Channel Leakage

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Abstract—Quantum computing threatens classical public key systems and has motivated NIST’s PQC standardization, and within this process HAWK has been submitted as a lattice-based signature scheme. While prior analyses focused on mathematical security, the consequences of side-channel leakage in practical HAWK implementations are less systematically understood. We address this gap by formalizing discrete Gaussian sampler leakage as HAWK with Hint and studying three leakage categories: (i) full coefficient recovery, (ii) sign-only recovery, and (iii) noisy sign recovery. For each category, including (i) which has been covered in prior work, we design algebraic key recovery algorithms that derive explicit linear relations between leaked information and the secret basis and solve the resulting structured linear systems in polynomial time.

For HAWK-1024, we obtain the following results under the stated threat models. With full coefficient leakage, a single signature suffices for secret key recovery by directly solving the induced linear system, improving prior work that required two signatures under the same conditions. With sign-only leakage, the secret key is recovered using 14 signatures in approximately 100 seconds. With noisy sign leakage, we achieve key recovery using 400 signatures within about 30 seconds at a 10% bit-error rate, and still succeed with roughly 7,000 signatures within about one minute even when the bit-error rate increases to 40%. We implement all attacks and empirically validate them, clarifying how measurement noise affects the required number of signatures.

1. Introduction

Quantum algorithms capable of efficiently solving the hard problems underlying traditional public-key cryptosystems such as RSA and ECC, together with rapid advances in quantum hardware, have made the need for quantum-resistant cryptographic systems increasingly urgent. To address this challenge, the National Institute of Standards and Technology (NIST) launched a public competition to standardize post-quantum cryptographic (PQC) schemes, focusing on both key encapsulation mechanisms (KEMs) and

digital signature algorithms (DSAs). In the KEM category, Kyber [1] and HQC [2] have been selected for standardization, while Dilithium [3] and Falcon [4] were chosen in the DSA category. In addition, NIST has initiated an extra evaluation round for DSAs, among which HAWK [5] has emerged as a promising candidate due to its balanced trade-off between performance and security.

Cryptographic algorithms are designed to be mathematically secure, but when implemented in hardware or software, they may leak additional information such as execution time, power consumption, or electromagnetic emanations. Exploiting such leakage, adversaries can mount side-channel attacks to recover secret keys. NIST has emphasized the importance of resistance against side-channel attacks throughout the PQC standardization process. Therefore, for newly proposed PQC schemes, it is critical to evaluate not only their mathematical security but also their implementation security in the presence of side-channel leakage.

Side-channel attacks have historically compromised otherwise secure cryptosystems: timing and power analysis have been used to break implementations of RSA and ECC, while electromagnetic analysis has exposed secrets in hardware devices. These examples highlight that mathematical security does not guarantee implementation security. In lattice-based post-quantum schemes, the discrete Gaussian sampler has emerged as a primary source of vulnerability, since its branching behavior and arithmetic operations induce data-dependent leakage [6], [7], [8], [9], [10], [11]. Evaluating the resilience of PQC schemes against such implementation attacks is therefore as critical as assessing their underlying hardness assumptions.

Recent studies confirm that samplers are a recurring point of weakness in PQC implementations. For lattice-based signatures, discrete Gaussian sampling and rejection-based routines have been exploited through timing, power, and EM analysis. Several studies on Falcon, for example, demonstrated that side-channel observations of signs, table accesses, and rejection behavior can be leveraged to recover the secret key [6], [8], [12], [13]. These results make it clear that, independent of asymptotic hardness guarantees, PQC schemes that rely on samplers must be carefully assessed against implementation-level attacks.

For HAWK, where the discrete Gaussian sampler is a

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core building block, systematically modeling and analyzing such leakages is therefore essential. The only notable prior work to date examined the side-channel vulnerability of HAWK’s sampler, demonstrating experimentally that secret key recovery is possible by combining partial information such as sign, parity, and increment leaked during sampling [14]. In this process, the authors also raised an open problem regarding how side-channel information could be incorporated into the lattice reduction framework when the secret key is represented in the form of HAWK’s Gram matrix Q . However, this study was conducted under low-noise laboratory conditions and did not consider more realistic, high-noise scenarios. Consequently, while it suggested the feasibility of such attacks, it fell short of establishing their practicality in real-world environments.

In realistic high-noise environments, as reported in [14], the identification of partial information observed during sampler execution can become significantly more difficult. Even when an adversary is able to directly target the intended leakage points, noise may introduce false positives and misdetections, suggesting that conventional recovery procedures that simply combine leakage signals may not be as effective as expected. For example, among the critical leakage sources such as sign, parity, and increment, some may be entirely unrecoverable under noisy conditions, and even if access to all three is possible, the accuracy of their recovery may be far from perfect, substantially degrading the reliability of the accumulated information. Therefore, research is needed to systematically quantify how much meaningful leakage can still be obtained under varying noise levels and error rates, and to rigorously define the feasibility and limitations of key recovery in such constrained settings.

This raises the central question:

How resilient is HAWK to key recovery attacks when adversaries obtain partial side-channel information?

1.1. Contribution

To address this question, we study HAWK in the presence of side-channel leakage and develop a structured framework for analyzing its security in the presence of partial leakage. Our findings reveal that HAWK is significantly more vulnerable to side-channel attacks than previously recognized, underscoring the urgent need for robust countermeasures in secure implementations. The main contributions of this work are as follows:

- **Formalization of HAWK with Hint.** We introduce the first formal framework for modeling side-channel information in HAWK. Rather than treating leakages as ad hoc hints, we define HAWK with Hint to capture the adversary’s partial knowledge of the Gaussian sampler output. Within this framework, we classify leakage into three categories: (i) full coefficient recovery, (ii) sign information, and (iii) partial or noisy sign information. This formalization provides a clear structure for analyzing HAWK under realistic leakage scenarios.

- **New algebraic identities solving the hint-integration problem.** We uncover previously unnoticed algebraic relations linking the signs used in HAWK’s discrete Gaussian sampling to the secret basis B . These direct linear yield constraints that resolve the open hint-integration problem from prior work [14]. These identities allow sign information to be transformed into exact algebraic constraints, serving as the key technical tool for our recovery algorithms.
- **Category-specific key recovery algorithms.** For each leakage category, we design explicit key recovery algorithms that run in polynomial time and exploit new algebraic relations between the discrete Gaussian sampler output and the secret basis:
 - *Category I (full coefficients):* We prove that, under full coefficient leakage, a single signature suffices for exact algebraic key recovery by directly solving the induced linear system, improving prior work that required two signatures.
 - *Category II (sign-only information):* We show that exact sign leakage alone suffices for polynomial-time key recovery, by reducing the problem to a structured linear program that can be efficiently solved by an interior-point method.
 - *Category III (noisy signs):* We design a polynomial-time key recovery algorithm that remains effective under independent bit-flip noise $p < \frac{1}{2}$, and theoretically characterize how the required number of signatures grows with the noise level.
- **Implementation and validation.** We implement all attacks and evaluate them on standard HAWK parameter sets, with detailed experiments for HAWK-1024.
 - *Category I:* A single HAWK-1024 signature suffices for key recovery and the corresponding linear system is solved essentially instantaneously.
 - *Category II:* Under exact sign-only leakage, our LP-based attack recovers the HAWK-1024 secret key using 14 signatures in about 100 seconds.
 - *Category III:* With noisy sign leakage, our two-stage attack recovers the HAWK-1024 key using about 400 signatures in ≈ 30 seconds at a 10% bit-error rate, and still succeeds with roughly 7,000 signatures in about one minute at a 40% bit-error rate.

The latest version of our source code is available [here](#).

1.2. Related Work

Gaussian samplers have long been known to leak side-channel information. For CDT-based samplers, Kim et al. [7] recovered sampled values from a single power trace via Hamming-weight leakage. For Falcon’s Base Sampler, Guerreau et al. [6] used sampler leakage to filter signatures, and Zhang et al. [8] further exploited second-order leakage through spectral analysis of covariance matrices, showing that even single-bit sign leakage per sample enables practical key recovery. Broader studies on lattice signatures confirmed similar weaknesses [9], [10], [11]. More recently, Guerreau

et al. [14] demonstrated that HAWK’s Gaussian sampler allows near-perfect reconstruction of sample values from power traces, enabling full secret recovery from as few as two signatures.

Research on LWE has likewise examined how partial side-channel information, or hints, can be integrated into attacks for full secret recovery. Dachman-Soled et al. [15] formalized this framework, incorporating perfect, modular, approximate, and short-vector hints into primal lattice reduction and providing concrete security estimation. Later works broadened the scope. Wu et al. [16] applied Fourier analysis to exploit hints in dual attacks. May and Nowakowski [17] efficiently combined many perfect or mod- q hints and showed that with sufficient hints, LWE instances such as Kyber-512 can be broken in practice using only LLL reduction. Lu et al. [18] introduced independent hints on both secret and error and replaced LLL-based basis construction with fast matrix operations to further reduce the cost. Hermelink et al. [19] proposed distribution hints with BP and greedy solvers for noisy leakage, and Shao et al. [20] adapted these ideas to NTRU lattices, showing that partial coefficient leakage in Falcon-512 drastically lowers BKZ requirements and enables full key recovery within hours.

2. HAWK

In this section, we present the HAWK signature scheme and establish the algebraic notation that will be used throughout its description and analysis. Its design relies on NTRU-type bases, Hermitian forms, and trace-based inner products, which together provide the algebraic framework for both key generation and verification. We begin by introducing the necessary notation and conventions.

2.1. Notations

We work over the negacyclic ring

$$R_n = \mathbb{Z}[X]/(X^n + 1), \quad K_n = \mathbb{Q}[X]/(X^n + 1),$$

with $n = 2^m$. Bold lowercase (resp. uppercase) denote vectors (resp. matrices) over R_n ; indices start at 0.

Hermitian Involution and Trace Pairing. For $a(X) = \sum_{i=0}^{n-1} a[i]X^i \in R_n$, define

$$a^* := a(X^{-1}) \bmod (X^n + 1),$$

which is negacyclic coefficient reversal: $a^*[0] = a[0]$ and $a^*[i] = -a[n-i]$ for $1 \leq i < n$. For a matrix $M = (m_{ij}) \in R_n^{r \times r}$ we set $M^* := (m_{ji}^*)$. The normalized trace is $\text{Tr}(u) = \frac{1}{n} \text{Tr}_{K_n/\mathbb{Q}}(u)$, and we write

$$\langle f, g \rangle := \text{Tr}(f^*g), \quad \langle \mathbf{u}, \mathbf{v} \rangle := \text{Tr}(u_0^*v_0 + u_1^*v_1),$$

so that $\langle f, g \rangle = (f^*g)[0]$ extracts the constant coefficient of f^*g . For any Hermitian $Q \in R_n^{2 \times 2}$ we use the quadratic form

$$\|\mathbf{w}\|_Q^2 := \text{Tr}(\mathbf{w}^*Q\mathbf{w}).$$

Complex Embedding. Let $\zeta = e^{i\pi/n}$ be a primitive $2n$ -th root of unity. The canonical embedding $\sigma : K_n \rightarrow \mathbb{C}^n$ is defined by

$$\sigma(a) := (a(\zeta), a(\zeta^3), \dots, a(\zeta^{2n-1})) = (a(\zeta^{2k+1}))_{k=0}^{n-1}.$$

This embedding is $\mathbb{Q}$ -linear and extends naturally to R_n and to vectors/matrices over R_n by component-wise application. The Hermitian involution satisfies $\sigma(a^*) = \overline{\sigma(a)}$ (complex conjugation), and the trace pairing has the interpretation

$$\langle f, g \rangle = \frac{1}{n} \sum_{k=0}^{n-1} \sigma(f^*)_k \sigma(g)_k = \frac{1}{n} \sum_{k=0}^{n-1} \overline{\sigma(f)_k} \sigma(g)_k.$$

We also define $R_n^{\mathbb{R}} := \mathbb{R}[X]/(X^n + 1)$, which we use for real-valued operations such as computing norms and square roots during signature generation.

Vectorization and the Lift Operator. Fix the power basis $(1, X, \dots, X^{n-1})$ and let $\text{vec}(a) \in \mathbb{Z}^n$ be the coefficient vector of $a \in R_n$. Let $S \in \mathbb{Z}^{n \times n}$ denote the negacyclic shift matrix

$$S = \begin{pmatrix} 0 & 0 & \dots & 0 & -1 \\ 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & \dots & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \dots & 1 & 0 \end{pmatrix},$$

which represents multiplication by X in R_n , i.e., $\text{vec}(Xa) = S \text{vec}(a)$. For $a = \sum_{i=0}^{n-1} a[i]X^i \in R_n$, define the (negacyclic) convolution matrix

$$\text{Rot}(a) := \sum_{i=0}^{n-1} a[i] S^i \in \mathbb{Z}^{n \times n}.$$

Then for all $a, b \in R_n$,

$$\text{vec}(ab) = \text{Rot}(a) \text{vec}(b) = \text{Rot}(b) \text{vec}(a),$$

and $\text{Rot}(\cdot)$ is a ring homomorphism: $\text{Rot}(ab) = \text{Rot}(a) \text{Rot}(b)$, $\text{Rot}(1) = I_n$. Moreover, the negacyclic adjoint corresponds the transpose:

$$\text{Rot}(a)^\top = \text{Rot}(a^*).$$

For a 2×2 matrix $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in R_n^{2 \times 2}$ we define its lift to $\mathbb{Z}^{2n \times 2n}$ by

$$\text{Lift}(M) := \begin{pmatrix} \text{Rot}(a) & \text{Rot}(b) \\ \text{Rot}(c) & \text{Rot}(d) \end{pmatrix}.$$

Writing $\text{Vec}(\mathbf{u}) := (\text{vec}(u_0) \parallel \text{vec}(u_1)) \in \mathbb{Z}^{2n}$ for $\mathbf{u} = (u_0, u_1) \in R_n^2$, the lift linearizes ring multiplication:

$$\begin{aligned} \text{Vec}(M\mathbf{u}) &= \text{Lift}(M) \text{Vec}(\mathbf{u}), \\ \text{Lift}(MN) &= \text{Lift}(M) \text{Lift}(N). \end{aligned} \tag{1}$$

In particular, bilinear forms over R_n^2 can be evaluated via integer linear algebra on $\mathbb{Z}^{2n}$ by lifting all operands.

Discrete Gaussians over Parity Cosets. For $t \in \{0, 1\}$ and $\sigma > 0$, the one-dimensional discrete Gaussian on the coset $2\mathbb{Z} + t$ with standard deviation σ is the distribution on $\{2k + t : k \in \mathbb{Z}\}$ with probability mass function

$$\Pr[X = 2k + t] \propto \exp\left(-\frac{(2k + t)^2}{2\sigma^2}\right).$$

For a bit-vector $t = (t_1, \dots, t_{2n}) \in \{0, 1\}^{2n}$ we write

$$D_{2\mathbb{Z}^{2n} + t, \sigma}$$

for the product distribution whose coordinates are independent draws from the above one-dimensional laws on $2\mathbb{Z} + t_i$. Hence $D_{2\mathbb{Z}^{2n} + t, \sigma}$ is supported on the lattice coset $2\mathbb{Z}^{2n} + t$ and is spherically subgaussian: for an absolute constant $c > 0$,

$$\Pr\left[\|\mathbf{x}\|_2^2 > 2c(2n)\sigma^2\right] \leq e^{-\Omega(n)}.$$

In particular, bounds of the form $\|\mathbf{x}\|_2^2 \leq 8n\sigma^2$ hold except with negligible probability when σ is chosen as in the scheme parameters. When needed, we will write $D_{2\mathbb{Z}^{2n} + t, 2\sigma_{\text{sign}}}$ to reflect the specific signing parameter. For a single coefficient $X \sim D_{2\mathbb{Z} + t_i, 2\sigma_{\text{sign}}}$, we write $\mu_{\text{sign}} := \mathbb{E}[|X|]$ for the expected absolute value and note that $\text{Var}(X) = 4\sigma_{\text{sign}}^2$ follows from the coordinatewise standard deviation.

2.2. The HAWK Scheme

We briefly describe the algorithms of the HAWK signature scheme in terms of the notations introduced in [Section 2.1](#). The scheme consists of three algorithms: key generation, signing, and verification. For more details and further implementation, we refer the reader to the original HAWK paper [\[21\]](#).

Key Generation (HawkKeyGen). The algorithm produces a secret key $\text{sk} = B \in \text{GL}_2(R_n)$ and a public key

$$\text{pk} = (Q, \sigma_{\text{sign}}, \sigma_{\text{verify}}),$$

where $Q := B^*B$ is a positive-definite Hermitian matrix with determinant equal to 1.

To generate B , the algorithm first samples polynomials $f, g \in R_n$ from a centered binomial distribution with parameter η , so that in expectation

$$\|f\|_2^2, \|g\|_2^2 \approx \frac{n\eta}{2}.$$

Next, one computes polynomials $F, G \in R_n$ that satisfy the Bézout identity

$$fG - gF = 1,$$

and then defines

$$B = \begin{pmatrix} f & F \\ g & G \end{pmatrix}.$$

The construction is accepted only if (f, g) satisfy certain mild conditions—most importantly, that the constant term of q_{00}^{-1} in $\mathbb{Q}[X]/(X^n + 1)$ is smaller than a prescribed bound β_0 . If this requirement is not met, the sampling process is repeated until it succeeds.

Signature Generation (HawkSign). Given a message m and the public key pk , the signing algorithm outputs a signature $\sigma = (\text{salt}, s_1)$.

1) Compute

$$\begin{aligned} \text{hpub} &= \text{SHAKE256}(\text{pub}), \\ M &= \text{SHAKE256}(m \parallel \text{hpub}), \\ h &= (h_0, h_1) = \text{SHAKE256}(M \parallel \text{salt}) \in (R_n/2R_n)^2. \end{aligned}$$

2) Form

$$t \equiv Bh \pmod{2R_n} \leftrightarrow t \in \{0, 1\}^{2n},$$

and sample

$$\mathbf{x} = (x_0, x_1) \leftarrow D_{2\mathbb{Z}^{2n} + t, 2\sigma_{\text{sign}}}.$$

3) If $\|\mathbf{x}\|_2^2 > 8n\sigma_{\text{verify}}^2$, restart with a new salt.

4) Compute the raw value

$$\hat{\mathbf{w}} = B^{-1}\mathbf{x} = h + 2s, \quad s \in R_n,$$

with explicit relation $\hat{w}_1 = -gx_0 + fx_1$.

5) Apply the symmetry breaking. Let $i^* = \min\{i \in \{0, \dots, n-1\} : \hat{w}_1[i] \neq 0\}$, and set $\varkappa = \text{sgn}(\hat{w}_1[i^*]) \in \{\pm 1\}$. Define the canonical vector $\mathbf{w} := \varkappa \hat{\mathbf{w}}$, so that the first nonzero coefficient of w_1 is positive. Symmetry break refers to the procedure of selecting one among multiple symmetric candidate solutions—through randomness or sign determination—so that the secret key becomes uniquely identifiable.

6) Output the signature

$$\sigma = (\text{salt}, s_1), \quad s_1 = \frac{h_1 - w_1}{2} \in R_n.$$

Signature Verification (HawkVerify). Given a message m , the public key pk , and a signature $\sigma = (\text{salt}, s_1)$, the verifier accepts or rejects as follows.

1) Recompute

$$\begin{aligned} \text{hpub} &= \text{SHAKE256}(\text{pub}), \\ M &= \text{SHAKE256}(m \parallel \text{hpub}), \\ h &= (h_0, h_1) = \text{SHAKE256}(M \parallel \text{salt}). \end{aligned}$$

2) Reconstruct

$$\begin{aligned} w_1 &= h_1 - 2s_1, \\ s_0 &= \left\lceil \frac{h_0}{2} + \frac{q_{01}}{q_{00}} \left(\frac{h_1}{2} - s_1 \right) \right\rceil, \\ w_0 &= h_0 - 2s_0. \end{aligned}$$

and set $\mathbf{w} = (w_0, w_1)$.

3) Accept if and only if

$$\|\mathbf{w}\|_Q^2 = \text{T}(\mathbf{w}^*Q\mathbf{w}) \leq 8n\sigma_{\text{verify}}^2.$$

3. Side-Channel Leakage in HAWK’s DGS and the Induced Problem

The HAWK signature algorithm employs a Discrete Gaussian Sampler (DGS) to generate signatures. This sampling procedure is particularly sensitive to side-channel leakage, meaning that observable quantities such as power consumption or electromagnetic emanations can reveal information about the internal state.

These leakage can reveal partial information about the internal sample vectors, thereby making secret key recovery more feasible. In this section, we review the leakage phenomena observed in HAWK’s sampler, introduce a unified abstraction that we call HAWK with Hint, and recall the key recovery approach of [14] as a baseline.

3.1. Side-Channel Leakage in HAWK’s DGS

```

1 #Inputs: tab_hi, tab_lo (RCDTs), v (index), t
   (target), hi, lo (randomness)
2 #Outputs: x[v] (sample at index v), sn
   (squared norm of x)
3
4 uint32_t neg = -(uint32_t)(lo >> 63);
5 uint32_t pbit = (t[v >> 3] >> (v & 7)) & 1;
6 uint64_t p_odd = -(uint64_t)pbit;
7 uint32_t p_oddw = (uint32_t)p_odd;
8 uint32_t r = 0;
9 for (size_t i = 0; i < hi_len; i += 2) {
10     uint64_t tlo0 = tab_lo[i + 0];
11     uint64_t tlo1 = tab_lo[i + 1];
12     uint64_t tlo = tlo0 ^ (p_odd & (tlo0 ^
13     tlo1));
14     uint32_t cc = (uint32_t)((lo - tlo) >> 63);
15     uint32_t thi0 = tab_hi[i + 0];
16     uint32_t thi1 = tab_hi[i + 1];
17     uint32_t thi = thi0 ^ (p_oddw & (thi0 ^
18     thi1));
19     r += (hi - thi - cc) >> 31;
20 }
21 uint32_t hinz = (hi - 1) >> 31;
22 for (size_t i = hi_len; i < lo_len; i += 2) {
23     uint64_t tlo0 = tab_lo[i + 0];
24     uint64_t tlo1 = tab_lo[i + 1];
25     uint64_t tlo = tlo0 ^ (p_odd & (tlo0 ^
26     tlo1));
27     uint32_t cc = (uint32_t)((lo - tlo) > >
28     63);
29     r += hinz & cc;
30 }
31 r = (r << 1) - p_oddw;
32 r = (r ^ neg) - neg;
33 x[v] = (int8_t)*(int32_t *)&r;
34 sn += r * r;

```

Listing 1. Discrete Gaussian Sampler inside the HAWK reference C implementation

The DGS invoked in HAWK’s signing procedure performs a sequence of branching operations, table lookups, and arithmetic updates. Each of these steps produces characteristic side-channel leakage in power or electromagnetic traces, which can reveal information about the sampled coefficients. Following the analysis of [14], three types of leakage have been identified: sign, parity, and increment

leakage. Listing 1 shows the process of sampling a single coefficient through the DGS in the HAWK reference C implementation. In Lines 4 and 5, the sign and parity bits are determined. From Lines 9 to 26, the sampled random value is compared with each element of the predefined tables, where the table selection is dictated by the previously obtained parity. In Lines 27 and 28, the determined sign and parity are applied, and finally, in Line 30, the norm is computed. In this code, the leakage points corresponding to the three types are as follows.

3.1.1. Sign Leakage. Sign information leaks during three distinct stages: (i) the initial sign decision (Line 4), (ii) the application of the chosen sign to the sampled coefficient (Line 28), and (iii) the norm computation used in the rejection step (Line 30). In practice, power traces separate cleanly into two groups, so that a simple threshold classifier suffices to distinguish positive and negative coefficients with essentially perfect accuracy. An additional detail is the treatment of zeros: in Lines 4 and 28, coefficients that are zero but marked with a “negative sign” produce a distinguishable trace (a so-called *negative zero*). Since the norm computation consistently treats zero as nonnegative in Line 30, these negative zeros can be reliably identified at that stage, further reinforcing the accuracy of sign recovery.

3.1.2. Parity Leakage. Parity information (whether a coefficient is even or odd) is determined early in the sampling routine (Line 5) and reappears multiple times during subsequent table traversals. In the specific implementation studied in [14], each iteration of the first traversal yields two leakage points (Lines 12 and 16), while the second traversal yields one (Line 23), resulting in a total of 18 parity-related leakage points per coefficient. These repeated exposures produce two distinct classes of traces, corresponding to even and odd values, with amplitudes dominated by Hamming-weight differences in 64-bit registers. As a result, parity recovery is essentially error-free in practice.

3.1.3. Increment Leakage. The magnitude of the sampled coefficient is determined through a sequential comparison procedure based on a Recursive Cumulative Distribution Table (RCDT). A uniformly random value is compared against table entries in descending order, and the coefficient is incremented whenever the random value falls below the current threshold. Each comparison operation produces a distinct side-channel footprint, allowing an adversary to infer whether the magnitude was incremented at that iteration (Lines 13, 17). Although increment leakage is less robust than sign or parity leakage, it provides additional constraints on the distribution of coefficients and can be combined with other leakage to reduce the uncertainty about the sampled vector.

3.2. The HAWK with Hint problem

The leakage described in Section 3.1 (sign, parity, and increment information) demonstrate that the DGS in HAWK

reveals consistent, fine-grained clues about its internal outputs. We collectively refer to such side-channel-derived observations as hints. In our threat model, HAWK with Hint denotes the setting where an adversary obtains these clues in addition to the public transcript of the signing algorithm. The adversary's challenge is then twofold: (i) to extract hint information reliably from physical traces, and (ii) to leverage this information together with the public key to reconstruct the secret key.

The difficulty of this task depends critically on the form and quality of the hints, as well as on the number of signatures available. To formalize this problem, we adopt the following definition.

Definition 1. Let $B \in \text{GL}_2(R_n)$ be the secret key of HAWK and let $Q = B^*B$ be the public key. For each signing execution j , the sampler produces vectors $(\mathbf{x}^{(j)}, \mathbf{w}^{(j)}) \in R_n^2 \times R_n^2$ related by

$$\mathbf{x}^{(j)} = B \mathbf{w}^{(j)}, \quad \mathbf{w}^{(j)} = \mathbf{h}^{(j)} + 2s^{(j)}.$$

The adversary observes, in addition to the public transcript, side-channel information $I^{(j)}$ about the coefficients of $\mathbf{x}^{(j)}$.

Instance:

- the public key Q and transcript information from t signatures,
- side-channel observations $\mathcal{I} = \{I^{(j)}\}_{j=1}^t$, where each $I^{(j)}$ conveys additional information about $\text{Vec}(\mathbf{x}^{(j)})$. Specifically, the first n indices correspond to coefficients of $x_0^{(j)}$ and the last n indices to those of $x_1^{(j)}$.

Goal: Recover the secret basis B of HAWK.

Depending on the leakage quality, we distinguish three categories:

Category I: Full coefficient recovery.

Each coefficient of $\mathbf{x}$ is recovered exactly, reducing key recovery to a direct linear-algebra problem.

Category II: Sign information.

Only the sign of each coefficient is obtained. This constrains each coefficient of $\mathbf{x}$ but does not uniquely determine it.

Category III: Noisy sign information.

The adversary learns sign information for only a subset of coefficients, or receives sign bits corrupted by errors. This weaker setting requires combining imperfect leakage with algebraic structure to attempt recovery.

These categories abstract the spectrum of leakage scenarios for HAWK and provide the framework for the recovery algorithms developed in the next sections.

3.3. key recovery from Two Signatures

The side-channel analysis of HAWK by Guerreau et al. [14] showed that leakage from the discrete Gaussian sampler expose nearly complete information about the sampled vector $\mathbf{x}$, enabling full key recovery. This setting corresponds to Category I in our framework, where two signatures provide exact coefficient values (up to symmetry), allowing direct algebraic reconstruction of the secret basis.

Recovery strategy. Let $B \in R_n^{2 \times 2}$ denote the secret basis, and let each signature produce vectors $\mathbf{w} \in R_n^2$ and $\mathbf{x} = B\mathbf{w} \in R_n^2$. Given two signatures $(\mathbf{x}, \mathbf{w})$ and $(\mathbf{x}', \mathbf{w}')$, define

$$X = [\mathbf{x} \ \mathbf{x}'], \quad W = [\mathbf{w} \ \mathbf{w}'], \quad \text{so that} \quad B = XW^{-1}.$$

Hence, two linearly independent pairs suffice for exact key recovery.

Experimental leakage and complexity. In the experiments of [14], about 98% of coefficients were recovered exactly, with the remainder limited to $\{\pm 4, \pm 6\}$. Enumerating assignments consistent with $\|\mathbf{x}\|^2 = \sum_i x[i]^2$ reduces the residual space below 2^{30} (about 2^{18} in practice). For $n = 1024$, computing $B = XW^{-1}$ requires roughly 14 polynomial multiplications, i.e., fewer than 2^{55} elementary operations.

In summary, the prior work demonstrates that in the strongest leakage model, the secret basis B can be efficiently recovered from only two signatures. This case serves as the baseline for our subsequent analysis of weaker and noisier leakage categories.

4. Key Recovery Analysis under HAWK with Hint

Unless stated otherwise, the proofs of all lemmas and theorems in this section are deferred to Appendix A.

4.1. Algebraic Preliminaries

We first summarize the algebraic relations that hold for any HAWK instance and that will be referenced in the key recovery analysis. Let $B = \begin{bmatrix} f & F \\ g & G \end{bmatrix} \in \text{GL}_2(R_n)$ with $\det B = 1$, and define the public matrix $Q = B^*B = \begin{bmatrix} q_{00} & q_{01} \\ q_{10} & q_{11} \end{bmatrix}$. We recall the notion of a Gram matrix. For a basis matrix B with column vectors, the matrix B^*B encodes all pairwise inner products of these columns and is Hermitian positive definite. In our setting, the public matrix Q satisfies $Q = B^*B$, so Q is a Gram matrix and inherits these properties. In particular, it satisfies the Gram relation $BQ^{-1}B^* = I_2$, which we will repeatedly use. Since Q is Hermitian, $q_{10} = q_{01}^*$ and

$$q_{11} = \frac{|q_{01}|^2 + 1}{q_{00}}, \quad |q_{01}|^2 = q_{01}^* q_{10}. \quad (2)$$

During signing, for $\hat{\mathbf{w}} \in R_n^2$ we compute $\mathbf{x} = B\hat{\mathbf{w}}$ and output $\mathbf{w} = \varkappa \hat{\mathbf{w}}$ with $\varkappa \in \{\pm 1\}$. Writing $\mathbf{x} = (x_0, x_1)^T$ and $\mathbf{w} = (w_0, w_1)^T$, define

$$M_{\mathbf{x}} = \begin{bmatrix} x_0^* & x_1^* \\ -x_1 & x_0 \end{bmatrix}, \quad M_{\mathbf{w}} = \begin{bmatrix} (\mathbf{w}^*Q)_0 & (\mathbf{w}^*Q)_1 \\ -w_1 & w_0 \end{bmatrix}, \quad (3)$$

$$M_{\hat{\mathbf{w}}} = \begin{bmatrix} (\hat{\mathbf{w}}^*Q)_0 & (\hat{\mathbf{w}}^*Q)_1 \\ -\hat{w}_1 & \hat{w}_0 \end{bmatrix}.$$

Lemma 4.1. For $B, Q, \mathbf{x}, \hat{\mathbf{w}}$ defined above, the following relations hold:

$$M_{\hat{\mathbf{w}}} B^{-1} = M_{\mathbf{x}}, \quad M_{\hat{\mathbf{w}}} = M_{\mathbf{x}} B. \quad (4)$$

Lemma 4.2. For $\mathbf{w} = \varkappa \hat{\mathbf{w}}$ with $\varkappa \in \{\pm 1\}$, the corresponding M -matrices satisfy

$$M_{\mathbf{w}} = \varkappa M_{\hat{\mathbf{w}}}. \quad (5)$$

Lemma 4.3. The determinants of the matrices $M_{\mathbf{x}}, M_{\hat{\mathbf{w}}}$, and $M_{\mathbf{w}}$ are equal and nonzero:

$$\det(M_{\mathbf{w}}) = \det(M_{\hat{\mathbf{w}}}) = \det(M_{\mathbf{x}}) = (\mathbf{x}^* \mathbf{x}) \neq 0. \quad (6)$$

Lemma 4.1–Lemma 4.3 link the observable signing vectors $(\mathbf{x}, \mathbf{w})$ to the hidden basis B and form the algebraic foundation for the recovery algorithms presented in [Section 4.2–Section 4.5](#).

In the following, we first treat Category I in [Section 4.2](#). Then, in [Section 4.3](#), we address the sym-break issue that must be resolved for both Categories II and III. Finally, [Section 4.4](#) and [Section 4.5](#) describe Categories II and III, respectively.

4.2. Category I

We begin with the strongest leakage model, Category I, where the adversary observes both the sampled vector $\mathbf{x}$ and the corresponding canonical lattice vector $\mathbf{w}$ from a single signing execution. While previous work [\[14\]](#) required two independent pairs $(\mathbf{x}, \mathbf{w})$ to reconstruct the secret basis B , we show that a single pair already suffices. The only unavoidable ambiguity is a global sign: the recovery yields either B or $-B$, both of which induce the same public key $Q = B^* B$.

Information inferred from leakage. We recall that, in Category I, we assume that all coefficients of the vector $\mathbf{x}$ sampled from the DGS were fully and accurately recovered.

High-level idea. From the algebraic identities in [Section 4.1](#), the observed pair $(\mathbf{x}, \mathbf{w})$ satisfies

$$M_{\mathbf{w}} = \varkappa M_{\hat{\mathbf{w}}} = \varkappa M_{\mathbf{x}} B, \quad \varkappa \in \{\pm 1\}.$$

Here $M_{\mathbf{x}}$ is computed directly from $\mathbf{x}$ and $M_{\mathbf{w}}$ is computed from $(\mathbf{w}, Q)$. By [Lemma 4.3](#), both $M_{\mathbf{w}}$ and $M_{\mathbf{x}}$ are always invertible since $\det(M_{\mathbf{x}}) = \det(M_{\mathbf{w}}) = \mathbf{x}^* \mathbf{x} = \|\mathbf{x}\|^2 > 0$ in any valid HAWK signature. Therefore, we obtain an explicit candidate for the inverse secret basis:

$$\tilde{B} := M_{\mathbf{x}}^{-1} M_{\mathbf{w}} = \varkappa B.$$

Thus B is recovered up to a global sign by one matrix inversion and two matrix multiplications over R_n . Disambiguating the sign is information-theoretically impossible from $(Q, \mathbf{x}, \mathbf{w})$ alone because $(-B)^* (-B) = B^* B = Q$.

Result. Given a single observed pair $(\mathbf{x}, \mathbf{w})$ from one signing execution and the public key Q , the adversary can compute $\tilde{B} = M_{\mathbf{x}}^{-1} M_{\mathbf{w}}$ and thereby recover the secret basis up to sign, i.e., $\tilde{B} \in \{B, -B\}$. This procedure runs in polynomial time in n .

Theorem 4.4. Let $B \in \text{GL}_2(R_n)$ be the secret key of HAWK and let $Q = B^* B$ be the public key. Suppose that for one signing execution the adversary obtains both $\mathbf{x} \in R_n^2$ and the canonical vector $\mathbf{w} \in R_n^2$ where $\mathbf{w} = \varkappa \hat{\mathbf{w}}$ for some unknown $\varkappa \in \{\pm 1\}$ and $\mathbf{x} = B \hat{\mathbf{w}}$. Then the adversary can recover either B or $-B$ in polynomial time.

4.3. Handling sym-break

Before turning to Categories II–III, we first resolve the sample-wise sign symmetry introduced by HAWK’s sym-break. We leverage information from multiple signatures where $x^{(j)}$ denotes the sampled vector corresponding to the j -th signature. Each signature j involves an unknown sign $\varkappa^{(j)} \in \{\pm 1\}$ that must be recovered before aggregating information across signatures. The following spectral routine aligns these signs and is applicable to both Category II (exact sign) and Category III (noisy sign observations).

Information inferred from leakage. For each signature $j \in \{1, \dots, k\}$, the public transcript of HAWK reveals the canonical signature vector $\mathbf{w}^{(j)} = (w_0^{(j)}, w_1^{(j)}) \in R_n^2$ and the public key $Q = B^* B \in R_n^{2 \times 2}$. In addition, the side-channel attack yields partial information about the corresponding pre-canonical vector $\mathbf{x}^{(j)} = (x_0^{(j)}, x_1^{(j)}) \in R_n^2$.

We encode this leakage as $I^{(j)} = (I_0^{(j)}, I_1^{(j)}) \in R_n^2$, where each coefficient $I_i^{(j)}[k] \in \{\pm 1\}$ represents the observed sign and may be flipped independently with probability $p < 1/2$:

$$I_i^{(j)}[k] = \begin{cases} \text{sign}(x_i^{(j)}[k]) & \text{with probability } 1 - p, \\ -\text{sign}(x_i^{(j)}[k]) & \text{with probability } p. \end{cases}$$

For example, when the actual sample value is $x_i^{(j)}[k] = -3.2$, we have $\text{sign}(x_i^{(j)}[k]) = -1$, but with noise level $p = 0.1$ there is a 10% chance that the observed sign is flipped to $I_i^{(j)}[k] = 1$.

Thus, for [Algorithm 1](#) the per-signature input consists of the publicly known vector $\mathbf{w}^{(j)}$ and the side-channel sign pattern $I^{(j)}$, together with the public key Q , while $\mathbf{x}^{(j)}$ itself remains hidden. Category II corresponds to the special case $p = 0$ (exact signs), whereas Category III allows $p > 0$ (noisy signs).

The canonical signature vector $\mathbf{w}^{(j)} = (w_0^{(j)}, w_1^{(j)})$ is computable from the public transcript, and the unknown symmetry-breaking sign $\varkappa^{(j)} \in \{\pm 1\}$ relates $\mathbf{w}^{(j)} = \varkappa^{(j)} \hat{\mathbf{w}}^{(j)}$.

We define the observable feature vector $\ell^{(j)}$ and the theoretical feature vector $\mathbf{z}^{(j)}$, and present their associated properties as lemmas.

Definition 2 (Feature Vectors). Let $M_{\mathbf{w}}^{(j)}$ and $M_{\mathbf{x}}^{(j)}$ be the matrices defined in Equation (3). For each signature j , define

$$\begin{aligned}\ell^{(j)} &:= (M_{\mathbf{w}}^{(j)})^* \begin{pmatrix} (I_0^{(j)})^* \\ -I_1^{(j)} \end{pmatrix} \in \mathbb{R}_n^2, \\ \mathbf{z}^{(j)} &:= (M_{\mathbf{x}}^{(j)})^* \begin{pmatrix} (I_0^{(j)})^* \\ -I_1^{(j)} \end{pmatrix} =: \begin{pmatrix} \alpha^{(j)} \\ \beta^{(j)} \end{pmatrix} \in \mathbb{R}_n^2.\end{aligned}$$

Lemma 4.5. Let $\ell^{(j)}$ and $\mathbf{z}^{(j)}$ be as in Definition 2. Then:

1) **Algebraic relation.** Using Lemma 4.1 and Lemma 4.2, we have

$$\ell^{(j)} = \mathcal{K}^{(j)} B^* \mathbf{z}^{(j)}.$$

2) **Statistical moments.** The canonical embedding components are approximately distributed as

$$\sigma(\alpha^{(j)})_k \sim \mathcal{CN}(2n\mu, 8n^2\sigma_{\text{sign}}^2), \quad \sigma(\beta^{(j)})_k \sim \mathcal{CN}(0, 8n^2\sigma_{\text{sign}}^2)$$

for all $k \in \{0, \dots, n-1\}$, where $\mu = (1-2p)\mu_{\text{sign}}$.

High-level idea. Our goal is to align the unknown symmetry-breaking signs so that information can be aggregated across signatures. To this end, we work with the observable feature vectors $\{\ell^{(j)}\}_{j=1}^k$ from Definition 2 and build a Gram matrix whose entries encode pairwise correlations under the Q^{-1} inner product. The next lemma records the resulting spike structure, which is the key to the spectral sign-recovery procedure.

Lemma 4.6. Let $G_0 \in \mathbb{R}^{k \times k}$ have entries

$$(G_0)_{ij} := \mathbb{T}((\ell^{(i)})^* Q^{-1} \ell^{(j)}),$$

where $\ell^{(j)}$ and $\mathbf{z}^{(j)}$ are as in Definition 2. Then

$$(G_0)_{ij} = \mathcal{K}^{(i)} \mathcal{K}^{(j)} \mathbb{T}((\mathbf{z}^{(i)})^* \mathbf{z}^{(j)}).$$

In particular, under the noisy-sign model of Lemma 4.5 with flip rate $p < \frac{1}{2}$,

$$\mathbb{E}[G_0] = \sigma_0 I_k + \lambda \psi \psi^\top, \quad \psi = (\mathcal{K}^{(1)}, \dots, \mathcal{K}^{(k)})^\top,$$

with $\lambda = 4n^2\mu^2$ and $\sigma_0 = 16n^2\sigma_{\text{sign}}^2$, where $\mu = (1-2p)\mu_{\text{sign}}$.

Intuitively, $\mathbb{E}[G_0]$ consists of an isotropic noise part and a rank-one spike in the direction of the unknown sign vector ψ ; hence the top eigenvector of G_0 is expected to align with ψ .

Algorithm. Algorithm 1 summarizes the resulting spectral method for recovering the symmetry-breaking signs. The procedure constructs observable feature vectors, forms the Gram matrix implicitly via the lifted Q^{-1} , and reads off the signs from the leading eigenvector.

We express $G_0 = A \text{Lift}(Q^{-1}) A^\top$ where $A \in \mathbb{R}^{k \times 2n}$ and $\text{Lift}(Q^{-1}) \in \mathbb{R}^{2n \times 2n}$, avoiding explicit construction of the $k \times k$ matrix. The Lanczos method then computes the top eigenvector in time $O(Tkn)$, where T is the number of iterations, so the overall complexity is polynomial in both n and k .

Algorithm 1: Symmetry-Breaking Sign Recovery via Spectral Method

Input: $\{(\mathbf{w}^{(j)}, I^{(j)})\}_{j=1}^k$ where $\mathbf{w}^{(j)} \in \mathbb{R}_n^2$, $I^{(j)} = (I_0^{(j)}, I_1^{(j)}) \in \mathbb{R}_n^{2 \times 2}$ with coefficients in $\{\pm 1\}$; public key $Q \in \mathbb{R}_n^{2 \times 2}$

Output: Symmetry-breaking signs $\{\mathcal{K}^{(j)}\}_{j=1}^k$, aligned vectors $\{\hat{\mathbf{w}}^{(j)}\}_{j=1}^k$, and feature vectors $\{\ell^{(j)}\}_{j=1}^k$

```

1 for  $j \leftarrow 1$  to  $k$  do
2   Compute  $M_{\mathbf{w}}^{(j)}$  from  $\mathbf{w}^{(j)}$  and  $Q$  via Equation (3)
3    $\ell^{(j)} \leftarrow (M_{\mathbf{w}}^{(j)})^* \begin{pmatrix} (I_0^{(j)})^* \\ -I_1^{(j)} \end{pmatrix}$ 
4    $A_j \leftarrow \text{vec}(\ell^{(j)}) \in \mathbb{R}^{2n}$ 
5  $A \leftarrow [A_1; \dots; A_k] \in \mathbb{R}^{k \times 2n}$ 
6 Compute leading eigenvector  $\hat{\psi} \in \mathbb{R}^k$  of  $G_0 = A \text{Lift}(Q^{-1}) A^\top$  using the Lanczos method without forming  $G_0$  explicitly
7 for  $j \leftarrow 1$  to  $k$  do
8    $\mathcal{K}^{(j)} \leftarrow \text{sign}(\hat{\psi}_j)$ 
9    $\hat{\mathbf{w}}^{(j)} \leftarrow \mathcal{K}^{(j)} \mathbf{w}^{(j)}$ 
10 return  $\{\mathcal{K}^{(j)}\}_{j=1}^k$ ,  $\{\hat{\mathbf{w}}^{(j)}\}_{j=1}^k$ , and  $\{\ell^{(j)}\}_{j=1}^k$ 

```

Result. We now state the performance guarantee for the spectral sign-alignment procedure. It follows from Lemma 4.6 together with standard matrix concentration bounds and an application of the Davis–Kahan theorem.

Theorem 4.7. Given k pairs $\{(\mathbf{w}^{(j)}, I^{(j)})\}_{j=1}^k$ whose signs follow the noisy-sign model, if

$$k \geq \frac{8\sigma_{\text{sign}}^2 (\log k)^2}{n\mu^2} \left(1 + \frac{2\sigma_{\text{sign}}^2}{\mu^2}\right),$$

where $\mu = (1-2p)\mu_{\text{sign}}$, then the algorithm runs in polynomial time $T_1 = O(Tkn)$ and for each $j = 1, \dots, k$, with probability $1 - O(1/k)$, the output of Algorithm 1 satisfies

$$\text{sign}(\hat{\psi}_j) = \mathcal{K}^{(j)}.$$

This covers both Categories II and III.

4.4. Category II

Information inferred from leakage. After the symmetry-breaking step (Algorithm 1), we assume that the pre-canonical vectors $\{\hat{\mathbf{w}}^{(j)}\}_{j=1}^k$ have been correctly aligned. For each signature j , the public transcript of HAWK reveals the canonical vector $\mathbf{w}^{(j)} \in \mathbb{R}_n^2$ and the public key $Q = B^* B \in \mathbb{R}_n^{2 \times 2}$, while the side-channel leakage provides the sign information of the corresponding pre-canonical vector $\mathbf{x}^{(j)} \in \mathbb{R}_n^2$.

In Category II, this leakage contains only exact signs of all coefficients, encoded as $I^{(j)} = (I_0^{(j)}, I_1^{(j)}) \in \mathbb{R}_n^2$ with $I_i^{(j)}[k] = \text{sign}(x_i^{(j)}[k]) \in \{\pm 1\}$. This corresponds to the noiseless case ($p = 0$) of the noisy-sign model in [Section 4.3](#), providing exact sign recovery for all coefficients.

High-level idea. From the aligned pre-canonical vectors $\hat{\mathbf{w}}^{(j)}$ and the algebraic relation in [Lemma 4.1](#), each observed sign pattern yields linear constraints on $\mathbf{z}_{\text{key}} = \text{Vec}(G, -g)$, the first column of B^{-1} . Crucially, these constraints correspond to absolute values of coefficients sampled from a discrete Gaussian, which are non-negative integers.

The key geometric insight is that the constraints define parallel hyperplanes orthogonal to $\mathbf{z}_{\text{key}}$, indexed by $k \in \{0, 1, 2, \dots\}$. Among these, the constraints corresponding to zero coefficients lie precisely on the hyperplane $k = 0$ passing through the origin. When sufficiently many such zero-coefficient constraints are in general position in the perpendicular subspace, they collapse the feasible cone into a single ray in the direction of $\mathbf{z}_{\text{key}}$, thereby uniquely determining the secret basis.

We formalize this constraint structure in [Definition 3](#) and [Property 1](#), followed by the main recovery theorem.

Definition 3. For each signature $j \in \{1, \dots, k\}$, let $\mathbf{v}^{(j)} = \text{Vec}((I_0^{(j)})^*, -I_1^{(j)}) \in \{\pm 1\}^{2n}$ encode the observed sign pattern. Define the constraint matrix

$$A^{(j)} = \text{Diag}(\mathbf{v}^{(j)}) \cdot \text{Lift}(M_{\hat{\mathbf{w}}^{(j)}}) \in \mathbb{R}^{2n \times 2n}.$$

The aggregate constraint matrix $A \in \mathbb{R}^{m \times 2n}$ (where $m = 2nk$) is the vertical stack of all constraint matrices:

$$A = \begin{bmatrix} A^{(1)} \\ \vdots \\ A^{(k)} \end{bmatrix}.$$

We denote the rows of A by $\mathbf{a}_i \in \mathbb{R}^{2n}$ for $i = 1, \dots, m$.

Property 1. Let $\mathbf{z}_{\text{key}} = \text{Vec}(G, -g) \in \mathbb{Z}^{2n}$ denote the first column of B^{-1} . Then

$$A\mathbf{z}_{\text{key}} \succeq \mathbf{0}.$$

Moreover, each entry $(A\mathbf{z}_{\text{key}})_i = \langle \mathbf{a}_i, \mathbf{z}_{\text{key}} \rangle$ equals the absolute value of a coefficient of $\mathbf{x}^{(j)}$ and is therefore a non-negative integer.

To find a feasible solution satisfying $A\mathbf{z} \succeq \mathbf{0}$, we transform the constraint system into a linear program and solve it using the Interior Point Method (IPM). The complete LP formulation, including the objective function, normalization constraints, and the detailed IPM algorithm, is provided in [Appendix B](#).

Algorithm. The recovery proceeds in two stages: first, we obtain $\hat{\mathbf{w}}^{(j)}$ for each signature via the symmetry-breaking algorithm ([Algorithm 1](#)); second, we construct the inequality constraints $A^{(j)}\mathbf{z}_{\text{key}} \succeq \mathbf{0}$ from the aligned pre-canonical vectors and sign patterns, formulate the linear program (see [Appendix B](#)), and solve it using the Interior Point Method

Algorithm 2: Category II: Key Recovery via Linear Program

Input: Public key $Q = B^*B \in \mathbb{R}_n^{2 \times 2}$; aligned pre-canonical vectors $\{\hat{\mathbf{w}}^{(j)}\}_{j=1}^k$ and exact sign patterns $\{I^{(j)}\}_{j=1}^k$ from [Algorithm 1](#)

Output: Secret basis B up to a global sign

```

1 for  $j = 1$  to  $k$  do
2   Compute  $M_{\hat{\mathbf{w}}^{(j)}}^{(j)}$  from  $\hat{\mathbf{w}}^{(j)}$  and  $Q$  via
   Equation (3)
3   Compute sign vector
    $\mathbf{v}^{(j)} \leftarrow \text{Vec}((I_0^{(j)})^*, -I_1^{(j)})$ 
4   Build constraint matrix
    $A^{(j)} \leftarrow \text{Diag}(\mathbf{v}^{(j)}) \cdot \text{Lift}(M_{\hat{\mathbf{w}}^{(j)}}^{(j)})$ 
5 Stack all constraints into  $A \in \mathbb{R}^{m \times 2n}$  with
    $m = 2nk$ 
6  $\mathbf{z}^* \leftarrow \text{Algorithm 4}(A)$ 
7  $\text{sol}_m \leftarrow \min\{|\mathbf{z}_i^*| : |\mathbf{z}_i^*| > \varepsilon\}$ 
8  $\mathbf{z}_{\text{key}} \leftarrow \text{round}(\mathbf{z}^*/\text{sol}_m)$ 
9 Write  $\mathbf{z}_{\text{key}} = \text{Vec}(G, -g)$ 
10  $(f, F) \leftarrow (G^*, -g^*)Q$  (first row of
    $B = (B^{-1})^*Q$ )
11 return  $B$ 
```

to recover the ray whose direction equals $\mathbf{z}_{\text{key}}$, from which the basis B is reconstructed.

The time complexity of the naive formulation is $O(k^3n^3)$ due to solving an $(2n + 2kn + 1) \times (2n + 2kn + 1)$ system; however, by exploiting the structure of the weight matrix W in the IPM iteration, the system can be compressed to $(2n + 1) \times (2n + 1)$, reducing the complexity to $O(kn^3)$. A detailed algorithmic description and complexity analysis are provided in [Appendix B](#).

Result. We now state the performance guarantee for the linear programming approach in Category II. The success probability depends on the number of zero-coefficient constraints being in general position, which is analyzed using Wendel's theorem on random hyperplane arrangements.

Theorem 4.8. Let B be the secret key and $Q = B^*B$ the public key of HAWK. Given k pairs $\{(\hat{\mathbf{w}}^{(j)}, I^{(j)})\}_{j=1}^k$ with exact sign observations ($p = 0$), the success probability of recovering B in time $T_2 = O(kn^3)$ is determined by whether the observed hyperplanes are in general position. Specifically,

$$\Pr[\text{key recovery} \mid k] = \mathbb{E}_{m \sim \text{Bin}(2nk, p_0)}[P_{2n-1}(m)]$$

where $p_0 = \frac{1}{2} \Pr_{Z \sim D_{\mathbb{Z}, \sigma_{\text{sign}}}}[Z = 0]$ and

$$P_d(m) := 1 - 2^{-m+1} \sum_{j=0}^{d-1} \binom{m-1}{j}$$

is the probability that m random hyperplanes through the origin in $\mathbb{R}^d$ are in general position (Wendel [22]).

For HAWK parameters with exact sign observations ($p = 0$), [Theorem 4.7](#) guarantees successful symmetry-breaking sign recovery.

For the success probability, $P_d(m)$ converges rapidly to 1 when $m \geq 2d$. In particular, for the dimension $d = 2n - 1$ relevant to our setting, once the number of zero-coefficient constraints m exceeds roughly $2(2n - 1)$, recovery becomes highly likely. Since $m \sim \text{Bin}(2kn, p_0)$ with expectation $\mathbb{E}[m] = 2knp_0$, key recovery is expected when $k \gtrsim 2/p_0$.

The total time complexity combines the spectral alignment phase (with T Lanczos iterations) and the linear programming phase, yielding $T_1 + T_2 = O(Tkn + kn^3)$.

4.5. Category III

Information inferred from leakage. Following the symmetry-breaking step ([Algorithm 1](#)), we assume that the pre-canonical vectors $\{\hat{\mathbf{w}}^{(j)}\}_{j=1}^k$ are correctly aligned. In Category III, the side-channel leakage provides noisy sign information of the sampled vectors $\mathbf{x}^{(j)} \in R_n^2$. For each signature j , the public transcript reveals $\mathbf{w}^{(j)} \in R_n^2$ and $Q = B^*B$, while the leakage is represented as $I^{(j)} = (I_0^{(j)}, I_1^{(j)}) \in R_n^2$ with coefficients $I_i^{(j)}[k] \in \{\pm 1\}$ flipped independently with probability $p < 1/2$:

$$I_i^{(j)}[k] = \begin{cases} \text{sign}(x_i^{(j)}[k]) & \text{with probability } 1 - p, \\ -\text{sign}(x_i^{(j)}[k]) & \text{with probability } p. \end{cases}$$

This generalizes the noiseless case ($p = 0$) of Category II, representing the full noisy-sign model introduced in [Section 4.3](#).

High-level idea. From the aligned pre-canonical vectors $\hat{\mathbf{w}}^{(j)}$ obtained via the spectral method in [Section 4.3](#), we have recovered the symmetry-breaking signs $\{\mathcal{X}^{(j)}\}_{j=1}^k$. By the algebraic relation in [Lemma 4.5](#), each signature yields

$$\mathcal{X}^{(j)} \ell^{(j)} = B^* \mathbf{z}^{(j)},$$

where $\ell^{(j)}$ is computable from public information and $\mathbf{z}^{(j)} = (\alpha^{(j)}, \beta^{(j)})^T$ has statistical properties $\mathbb{E}[\alpha^{(j)}] = c$, $\mathbb{E}[\beta^{(j)}] = 0$ with $c = \Theta(n(1 - 2p))$.

The key insight is that if we sum the theoretical aggregates across all signatures and normalize by the aggregate norm, the result concentrates near the direction $(1, 0)$:

$$\frac{\mathbf{z}_\Sigma}{\|\mathbf{z}_\Sigma\|} \approx \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

where $\mathbf{z}_\Sigma = \sum_{j=1}^k \mathbf{z}^{(j)}$. This reveals the first column of B^* since

$$\frac{\ell_\Sigma}{\|\mathbf{z}_\Sigma\|} = B^* \frac{\mathbf{z}_\Sigma}{\|\mathbf{z}_\Sigma\|} \approx B^* \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} f^* \\ F^* \end{pmatrix}.$$

Crucially, the normalization factor $\|\mathbf{z}_\Sigma\|^2$ can be computed from public information via the identity $(\ell_\Sigma)^* Q^{-1} \ell_\Sigma = (\mathbf{z}_\Sigma)^* \mathbf{z}_\Sigma$, enabling slot-wise normalization in the canonical embedding and coefficient-wise recovery via rounding.

We formalize the aggregation procedure and statistical properties in the following definitions and lemmas.

Definition 4. For signatures $j \in \{1, \dots, k\}$ with aligned symmetry-breaking signs $\{\mathcal{X}^{(j)}\}_{j=1}^k$ and feature vectors $\{\ell^{(j)}, \mathbf{z}^{(j)}\}_{j=1}^k$ from [Definition 2](#), define the **aligned aggregate feature**

$$\ell_\Sigma := \sum_{j=1}^k \mathcal{X}^{(j)} \ell^{(j)} \in R_n^2$$

and the **theoretical aggregate**

$$\mathbf{z}_\Sigma := \sum_{j=1}^k \mathbf{z}^{(j)} = \begin{pmatrix} \alpha_\Sigma \\ \beta_\Sigma \end{pmatrix} = \begin{pmatrix} \sum_{j=1}^k \alpha^{(j)} \\ \sum_{j=1}^k \beta^{(j)} \end{pmatrix} \in R_n^2.$$

Lemma 4.9. The aggregate vectors from [Definition 4](#) satisfy the following properties.

1) **Algebraic relations.** Using the relation from [Lemma 4.5](#) and the Gram relation $BQ^{-1}B^* = I_2$, we have

$$\ell_\Sigma = B^* \mathbf{z}_\Sigma \quad \text{and} \quad (\ell_\Sigma)^* Q^{-1} \ell_\Sigma = (\mathbf{z}_\Sigma)^* \mathbf{z}_\Sigma.$$

2) **Normalization property.** Under the noisy-sign model with independent flip rate $p < \frac{1}{2}$, the normalized aggregate $\mathbf{z}_\Sigma / \|\mathbf{z}_\Sigma\|$ concentrates near the direction $(1, 0)^T$ with error variance $\Theta(1/k)$ per component.

Algorithm. The recovery proceeds in two stages. First, we apply the spectral method from [Algorithm 1](#) to recover the symmetry-breaking signs $\{\mathcal{X}^{(j)}\}_{j=1}^k$ by constructing observable feature vectors $\ell^{(j)}$, forming the Gram matrix G_0 with Q^{-1} inner products, and extracting signs from the top eigenvector. Second, we aggregate the aligned features $\ell_\Sigma = \sum_{j=1}^k \mathcal{X}^{(j)} \ell^{(j)}$, compute the normalization factor $\|\mathbf{z}_\Sigma\|^2$ from the identity $(\ell_\Sigma)^* Q^{-1} \ell_\Sigma = (\mathbf{z}_\Sigma)^* \mathbf{z}_\Sigma$, perform slot-wise normalization in the canonical embedding to approximate $f^*(\zeta^s)$, and recover f via coefficient-wise rounding, from which the basis B is reconstructed.

The time complexity is $O(Tkn)$ for Stage 1, where T is the number of Lanczos iterations (typically $T = O(1)$), and $O(kn + n \log n)$ for Stage 2, yielding overall polynomial-time complexity in both n and k . A detailed algorithmic description is given in [Algorithm 3](#).

Result. Combining the eigenvector concentration, Gaussian tail bounds, and moment relations, we obtain the following guarantee.

Theorem 4.10. Suppose the symmetry-breaking signs $\{\mathcal{X}^{(j)}\}_{j=1}^k$ have been correctly recovered. If

$$k \geq \frac{12C\eta\sigma_{\text{sign}}^2 \log n}{\mu^2},$$

where $\mu = (1 - 2p)\mu_{\text{sign}}$, then the aggregation and slot-wise normalization procedure recovers f^* (and hence the secret basis B) in time $T_3 = O(kn + n \log n)$ via coefficient-wise rounding, with probability at least $1 - n^{-1 - \Omega(C)}$, where $C > 0$ is an absolute constant controlling the confidence level.

Algorithm 3: Category III: Key Recovery from Noisy Sign Information

Input: Public key $Q = B^*B \in R_n^{2 \times 2}$;
symmetry-breaking signs
 $\{\mathcal{X}^{(j)}\}_{j=1}^k \subset \{\pm 1\}$ and feature vectors
 $\{\ell^{(j)}\}_{j=1}^k$ from Algorithm 1

Output: Secret basis B up to a global sign

```

1  $\ell_\Sigma \leftarrow \mathbf{0} \in R_n^2$ 
2 for  $j \leftarrow 1$  to  $k$  do
3    $\ell_\Sigma \leftarrow \ell_\Sigma + \mathcal{X}^{(j)} \ell^{(j)}$ 
4  $T \leftarrow (\ell_\Sigma)^* Q^{-1} \ell_\Sigma \in R_n$ 
5 for each embedding slot  $s \in \{1, 3, \dots, 2n-1\}$  do
6    $r_s \leftarrow \sqrt{T(\zeta^s)} \in \mathbb{R}_{>0}$ 
7    $\hat{f}(\zeta^s) \leftarrow (\ell_\Sigma^{(0)})^* (\zeta^s) / r_s \in \mathbb{C}$ 
8 Apply inverse canonical embedding to obtain
    $\hat{f} \in K_n$ 
9  $f \leftarrow$  round each coefficient of  $\hat{f}$  to the nearest
   integer
10 Recover  $F, g, G$  from  $f$  using the relations
    $Q = B^*B$ 
11 return  $B$ 

```

Together with Theorem 4.7, this implies that whenever k simultaneously satisfies the spectral-alignment condition and the lower bound in Theorem 4.10, the two-stage Category III procedure recovers B in time $T_1 + T_3 = O(Tkn + kn + n \log n)$ with overwhelming probability in both n and k .

5. Implementation

We experimentally validate the proposed key recovery methodology by performing side-channel analyses on a HAWK reference implementation. The experiments leverage both low-noise power traces collected with the ChipWhisperer platform and traces with artificially injected noise to emulate real-world measurement conditions. The objectives are twofold: (i) to characterize how sign, parity, and increment leakages evolve as the noise level increases, and (ii) to empirically determine the amount of side-channel information required for practical key recovery under each leakage category. Furthermore, using the identified HAWK with Hint under various noise conditions, we perform secret-key recovery attacks on the actual HAWK implementation. For each noise scenario, we select the most appropriate approach from the attack categories proposed earlier.

5.1. Side-Channel Analysis on HAWK’s DGS

Experimental setup. All power traces were collected using the ChipWhisperer-Husky platform targeting an ARM Cortex-M4-based STM32F3 microcontroller. Following the methodology of prior work [14], the `SamplerSign` code

was modified to sample a single coefficient by isolating the corresponding loop in the HAWK-1024 reference implementation. Each execution that sampled one coefficient was recorded as a single power trace. Using 100 baseline traces obtained from single-coefficient sampling, we repeatedly injected independent Gaussian noise for each noise level and conducted a total of 1,000 experiments to analyze how side-channel leakages evolve across different noise conditions.

Noise-injected traces. To emulate realistic noise conditions, we injected artificial Gaussian noise into the power traces collected using the ChipWhisperer platform. We estimate σ_{trace} as a scalar baseline noise level by computing the per-sample variance across repeated acquisitions, selecting background windows unrelated to sampling activity, and averaging their variances. These segments correspond to regions not directly involved in the sampling process, and their variance levels are assumed to reflect the background measurement noise.

Based on σ_{trace} , we set the standard deviation of the injected noise, and for a chosen scaling factor α , the noisy trace at time index i is modeled as:

$$\text{Trace}_{\text{noisy}}[i] = \text{Trace}[i] + \mathcal{N}(0, (\alpha \cdot \sigma_{\text{trace}})^2). \quad (7)$$

In this paper, we considered scaling factors $\alpha \in \{1, 3, 5, 7, 10\}$ to systematically evaluate side-channel leakage variations under different noise levels.

Information extraction procedure. Our side-channel analysis pipeline consists of the following stages:

- 1) **Data acquisition:** we collected T traces per experiment per noise level. In our experiments we used $T = 100$.
- 2) **Preprocessing and alignment:** In our experiments, we repeatedly collected power traces targeting only the sampling of a single coefficient, and since the acquisition was performed using the ChipWhisperer platform, no additional trace alignment was required. However, when analyzing the full HAWK signing procedure, it becomes necessary to identify and segment the sampling intervals for each coefficient and perform alignment accordingly.
- 3) **Feature extraction:** To extract sign, parity, and increment information from the power traces, it is necessary to identify the points of interest (PoI) where such leakages occur. PoIs can be determined based on the structural characteristics of the HAWK implementation, for example, the intervals of repeated table comparisons as illustrated in Listing 1, the locations within each iteration that exhibit high variance across trace indices, and regions identified through SNR analysis. In our experiment, PoI identification was performed directly on the raw traces.
- 4) **Clustering:** For each PoI, k-means clustering with $k = 2$ was applied to extract the sign, parity, and increment information.
- 5) **Evaluation:** The clustering results obtained at each sign, parity, and increment leakage point were com-

TABLE 1. AVERAGE ACCURACY OF SIGN, PARITY, AND INCREMENT LEAKAGE UNDER DIFFERENT NOISE LEVELS (1,000 TRIALS).

α	Sign (acc.)	Parity (acc.)	Increment (acc.)
Original	100.0%	100.0%	100.0%
1	100.0%	100.0%	99.3%
3	99.9%	100.0%	89.8%
5	98.8%	99.9%	79.0%
7	96.8%	98.4%	72.2%
10	92.3%	93.2%	66.2%

pared with the ground-truth labels to compute the accuracy.

Information extraction accuracy. During the coefficient sampling phase of HAWK with Hint, three distinct forms of side-channel leakage are observed, each associated with sign, parity, or increment information. These leakages arise as different manifestations of how coefficient properties are reflected in the power traces and respond differently to injected noise. Representative leakage traces across various noise levels are presented in Appendix C.

Sign and parity leakages form two clearly separated clusters that distinguish negative from positive coefficients and odd from even coefficients. These separations provide stable structural cues about the internal state of the sampler and remain visually distinct even under considerable noise. This observation demonstrates that both sign and parity information are highly robust under noisy conditions and can be exploited reliably across a wide range of acquisition settings.

In contrast, increment leakage, which reflects whether an internal table comparison increases the coefficient value, appears much weaker. Because the Hamming-weight difference between increment and non-increment cases is relatively small, the separation between clusters collapses rapidly as noise increases. Beyond a certain noise level, this distinction becomes barely visible, indicating that increment-based leakage is difficult to exploit in realistic measurement environments.

Table 1 presents the average recovery accuracy for each leakage type as the injected noise parameter α varies. Following the clustering-based approach of prior work [14], we apply k -means clustering to the extracted leakage signals and average the success rates over 1,000 independent experiments for each noise level. In the absence of artificially injected noise, all three leakage types are recovered perfectly. As the noise level increases, the accuracies for sign and parity remain nearly perfect for low and moderate noise and reach about 93% even at the highest tested level. In contrast, the accuracy of increment leakage deteriorates much faster, dropping from near-perfect recovery at low noise to roughly two-thirds at $\alpha = 10$. This widening gap indicates that sign and parity information are resilient to noise, whereas increment leakage is fragile, making it unrealistic to achieve full coefficient recovery from all three leakage types simultaneously under practical noisy conditions.

TABLE 2. HAWK PARAMETER SETS AND STATISTICAL QUANTITIES.

Parameter Set	n	η	σ_{sign}	μ_{sign}	σ^2/μ^2	p_0
HAWK-256	256	2	1.010	1.578	0.409	0.197
HAWK-512	512	4	1.278	2.013	0.403	0.156
HAWK-1024	1024	8	1.299	2.047	0.403	0.154

Discussion of empirical findings.

- **Sign as practical primitive.** Since sign information can be extracted with consistently high accuracy across a wide range of noise levels, key recovery strategies that rely on sign bits (such as our Category II attack) remain viable in realistic environments. Although parity information is also extractable with high accuracy, our attacks focus exclusively on sign leakage.
- **Fragility of increment leakage.** The Hamming weight difference induced by increment decisions is relatively small and thus easily obliterated in the presence of Gaussian noise. Under high-noise conditions, exploiting this leakage becomes severely constrained, and in the worst case, the distinction of increment events in power traces may become ambiguous, rendering PoI identification impossible.

5.2. Key recovery attack under the HAWK with Hint

We validate the theoretical predictions from Section 4 through extensive simulations across all three HAWK parameter sets. Since Category I (full coefficient leakage) achieves deterministic key recovery from a single signature, we focus our experimental validation on Categories II and III, where statistical analysis is required. The recovery algorithms were implemented in Python using NumPy/SciPy, and executed on an AWS EC2 instance. Table 2 summarizes the three standardized HAWK instances and their derived statistical quantities, where $\mu_{\text{sign}} = \frac{1}{2}(\mathbb{E}[\|X_0\|] + \mathbb{E}[\|X_1\|])$ for coset Gaussians $X_i \sim D_{2\mathbb{Z}+i, 2\sigma_{\text{sign}}}$, and $p_0 = \frac{1}{2}\Pr[Z = 0]$ for $Z \sim D_{\mathbb{Z}, \sigma_{\text{sign}}}$ represents the zero-coefficient probability.

5.2.1. Category II: Sign leakage. Category II attacks assume the adversary obtains exact sign information for each coefficient of $\mathbf{x}^{(j)}$ across k signatures. Table 3 shows the empirical success rates over 100 independent trials. Practical key recovery requires only $k = 11$ signatures for HAWK-256 and $k = 14$ for HAWK-512/1024, achieving deterministic success at these thresholds. Empirical success rates consistently exceed theoretical predictions across all parameter sets: for instance, HAWK-256 achieves 56% at $k = 10$ versus the 39.8% theoretical prediction. This systematic improvement can be attributed to the negacyclic structure of HAWK’s polynomial ring $\mathbb{Z}[X]/(X^n + 1)$, where algebraic dependencies cause constraint vectors to exhibit greater angular spread than the random sphere model predicts.

TABLE 3. CATEGORY II SUCCESS RATES (100 TRIALS).

Parameter	Signatures k					
	10	11	12	13	14	15
HAWK-256	56%	100%	100%	100%	100%	100%
HAWK-512	0%	0%	1%	75%	100%	100%
HAWK-1024	0%	0%	0%	30%	100%	100%

TABLE 4. CATEGORY III SUCCESS RATE (100 TRIALS). EACH ENTRY SHOWS “REQUIRED SIGNATURES / SUCCESS RATE”.

Noise rate p	$C = 1$	$C = 2$	$C = 3$	$C = 4$
HAWK-256				
0.00	55 / 100%	110 / 100%	165 / 100%	220 / 100%
0.10	86 / 100%	172 / 100%	258 / 100%	344 / 100%
0.20	152 / 92%	304 / 100%	456 / 100%	608 / 100%
0.30	341 / 79%	682 / 100%	1,023 / 100%	1,364 / 100%
0.40	1,363 / 79%	2,726 / 100%	4,089 / 100%	5,452 / 100%
HAWK-512				
0.00	61 / 95%	122 / 100%	183 / 100%	244 / 100%
0.10	95 / 49%	190 / 100%	285 / 100%	380 / 100%
0.20	168 / 10%	336 / 99%	504 / 100%	672 / 100%
0.30	378 / 1%	756 / 85%	1,134 / 99%	1,512 / 100%
0.40	1,509 / 0%	3,018 / 81%	4,527 / 98%	6,036 / 100%
HAWK-1024				
0.00	67 / 6%	134 / 97%	201 / 100%	268 / 100%
0.10	105 / 0%	210 / 55%	315 / 98%	420 / 100%
0.20	187 / 0%	374 / 10%	561 / 76%	748 / 99%
0.30	419 / 0%	838 / 0%	1,257 / 49%	1,676 / 89%
0.40	1,675 / 0%	3,350 / 0%	5,025 / 31%	6,700 / 82%

Computationally, the runtime is dominated by linear programming solving, which scales as $O(n^3k)$ due to interior-point methods: HAWK-256 requires approximately 2 seconds and HAWK-1024 takes around 1 minute. Non-LP operations complete in under 1 second.

5.2.2. Category III: Noisy sign leakage. Category III attacks operate with noisy sign observations in which each sign bit is independently corrupted with probability p . The number of signatures is controlled through a confidence factor C , and larger values of C require proportionally more signatures to compensate for noise and dimensional effects. Table 4 summarizes our results based on one hundred independent trials for each setting. Each entry reports the number of signatures used followed by the empirical success rate, written in the format “required signatures / success rate”.

The results demonstrate that increasing the confidence factor C effectively compensates for both dimensional effects and noise-induced errors: at $C = 4$, HAWK-256 and HAWK-512 achieve perfect success across all tested noise levels up to $p = 0.40$, while HAWK-1024 achieves high success rates (at least 82%) even at the highest noise levels. Computationally, all attacks complete in under 1 minute, confirming that sign-based side-channel information poses a practical threat to HAWK’s security even under realistic noise conditions.

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Appendix A.

Proofs for Section 4

In this Appendix A, we provide detailed mathematical proofs for all lemmas and theorems stated in Section 4. Each result formalizes an algebraic or probabilistic relationship introduced in the main text and serves as the theoretical foundation for the proposed key recovery algorithms presented in Section 4.2–Section 4.5.

Lemma 4.1

For $B, Q, \mathbf{x}, \hat{\mathbf{w}}$ defined in Section 4.1, the matrices $M_{\mathbf{x}}$ and $M_{\hat{\mathbf{w}}}$ satisfy

$$M_{\hat{\mathbf{w}}}B^{-1} = M_{\mathbf{x}}, \quad M_{\hat{\mathbf{w}}} = M_{\mathbf{x}}B.$$

Proof. Recall that $Q = B^*B$, $\mathbf{x} = B\hat{\mathbf{w}}$ and

$$M_{\mathbf{x}} = \begin{bmatrix} x_0^* & x_1^* \\ -x_1 & x_0 \end{bmatrix}, \quad M_{\hat{\mathbf{w}}} = \begin{bmatrix} (\hat{\mathbf{w}}^*Q)_0 & (\hat{\mathbf{w}}^*Q)_1 \\ -\hat{w}_1 & \hat{w}_0 \end{bmatrix}.$$

Using $Q = B^*B$ and $\mathbf{x} = B\hat{\mathbf{w}}$,

$$\hat{\mathbf{w}}^*Q = \hat{\mathbf{w}}^*B^*B = (B\hat{\mathbf{w}})^*B = \mathbf{x}^*B.$$

Thus the first row of $M_{\hat{\mathbf{w}}}$ is $\mathbf{x}^*B$, and

$$((\hat{\mathbf{w}}^*Q)_0, (\hat{\mathbf{w}}^*Q)_1)B^{-1} = \mathbf{x}^*BB^{-1} = (x_0^*, x_1^*),$$

which is the first row of $M_{\mathbf{x}}$.

For the second row, since $\det B = 1$ and $\mathbf{x} = B\hat{\mathbf{w}}$, direct calculation gives $(-\hat{w}_1, \hat{w}_0)B^{-1} = (-x_1, x_0)$, the second row of $M_{\mathbf{x}}$. This proves $M_{\hat{\mathbf{w}}}B^{-1} = M_{\mathbf{x}}$.

Multiplying on the right by B gives $M_{\hat{\mathbf{w}}} = M_{\mathbf{x}}B$, as claimed. $\square$

Lemma 4.2

For $\mathbf{w} = \varkappa \hat{\mathbf{w}}$ with $\varkappa \in \{\pm 1\}$, the corresponding M -matrices satisfy

$$M_{\mathbf{w}} = \varkappa M_{\hat{\mathbf{w}}}.$$

Proof. By definition of the symmetry-breaking step,

$$\mathbf{w} = \varkappa \hat{\mathbf{w}}, \quad \varkappa \in \{\pm 1\},$$

so $\mathbf{w}^* = \varkappa \hat{\mathbf{w}}^*$ and

$$\mathbf{w}^*Q = \varkappa \hat{\mathbf{w}}^*Q.$$

Thus the first row of $M_{\mathbf{w}}$ is $\varkappa$ times the first row of $M_{\hat{\mathbf{w}}}$:

$$((\mathbf{w}^*Q)_0, (\mathbf{w}^*Q)_1) = \varkappa((\hat{\mathbf{w}}^*Q)_0, (\hat{\mathbf{w}}^*Q)_1).$$

For the second row,

$$(-w_1, w_0) = (-\varkappa \hat{w}_1, \varkappa \hat{w}_0) = \varkappa(-\hat{w}_1, \hat{w}_0).$$

Hence every row of $M_{\mathbf{w}}$ is multiplied by $\varkappa$, i.e.,

$$M_{\mathbf{w}} = \varkappa M_{\hat{\mathbf{w}}},$$

which proves the lemma. $\square$

Lemma 4.3

The determinants of the matrices $M_{\mathbf{x}}$, $M_{\hat{\mathbf{w}}}$, and $M_{\mathbf{w}}$ are equal and nonzero:

$$\det(M_{\mathbf{w}}) = \det(M_{\hat{\mathbf{w}}}) = \det(M_{\mathbf{x}}) = (\mathbf{x}^*\mathbf{x}) \neq 0.$$

Proof. By the 2×2 determinant formula, $\det(M_{\mathbf{x}}) = x_0^*x_0 + x_1^*x_1 = \mathbf{x}^*\mathbf{x} \neq 0$ since $\mathbf{x} \neq 0$.

From Lemma 4.1,

$$M_{\hat{\mathbf{w}}}B^{-1} = M_{\mathbf{x}} \implies \det(M_{\hat{\mathbf{w}}})\det(B^{-1}) = \det(M_{\mathbf{x}}).$$

Because $\det B = 1$, we have $\det(B^{-1}) = 1$, hence

$$\det(M_{\hat{\mathbf{w}}}) = \det(M_{\mathbf{x}}).$$

From Lemma 4.2, $M_{\mathbf{w}} = \varkappa M_{\hat{\mathbf{w}}}$ with $\varkappa \in \{\pm 1\}$, so

$$\det(M_{\mathbf{w}}) = \det(\varkappa M_{\hat{\mathbf{w}}}) = \varkappa^2 \det(M_{\hat{\mathbf{w}}}) = \det(M_{\hat{\mathbf{w}}}).$$

Combining,

$$\det(M_{\mathbf{w}}) = \det(M_{\hat{\mathbf{w}}}) = \det(M_{\mathbf{x}}) = \mathbf{x}^*\mathbf{x} \neq 0,$$

which proves the claim. $\square$

Theorem 4.4

Let $B \in \text{GL}_2(R_n)$ be the secret key of HAWK and let $Q = B^*B$ be the public key. Suppose that for one signing execution the adversary obtains both $\mathbf{x} \in R_n^2$ and the canonical vector $\mathbf{w} \in R_n^2$ where $\mathbf{w} = \varkappa \hat{\mathbf{w}}$ for some unknown $\varkappa \in \{\pm 1\}$ and $\mathbf{x} = B\hat{\mathbf{w}}$. Then the adversary can recover either B or $-B$ in polynomial time.

Proof. By Lemma 4.2 and Lemma 4.1, we have $M_{\mathbf{w}} = \kappa M_{\mathbf{x}} B$. Since $\det(M_{\mathbf{x}}) = \mathbf{x}^* \mathbf{x} \neq 0$ (Lemma 4.3), the adversary computes $\tilde{B} := M_{\mathbf{x}}^{-1} M_{\mathbf{w}} = \kappa B \in \{B, -B\}$. $\square$

Lemma 4.5

Let $\ell^{(j)}$ and $\mathbf{z}^{(j)}$ be as in Definition 2. Then:

- 1) **Algebraic relation.** Using Lemma 4.1 and Lemma 4.2, we have

$$\ell^{(j)} = \kappa^{(j)} B^* \mathbf{z}^{(j)}.$$

- 2) **Statistical moments.** The canonical embedding components are approximately distributed as

$$\sigma(\alpha^{(j)})_k \sim \mathcal{CN}(2n\mu, 8n^2\sigma_{\text{sign}}^2),$$

$$\sigma(\beta^{(j)})_k \sim \mathcal{CN}(0, 8n^2\sigma_{\text{sign}}^2).$$

for all $k \in \{0, \dots, n-1\}$, where $\mu = (1-2p)\mu_{\text{sign}}$.

Proof. (1) **Algebraic relation.** From Lemma 4.2 and Lemma 4.1, taking Hermitian transposes and multiplying by $((I_0^{(j)})^*, -I_1^{(j)})^T$ gives

$$\ell^{(j)} = \kappa^{(j)} B^* \mathbf{z}^{(j)}.$$

(2) **Statistical moments.** We compute the exact mean and variance in the canonical embedding.

Mean of $\sigma(\alpha^{(j)})_k$: By independence, only diagonal terms ($\ell = m$) contribute to the canonical embedding expansion. Under the noisy-sign model, $\mathbb{E}[I_0^{(j)}[\ell]x_0^{(j)}[\ell]] = \mu$, so $\mathbb{E}[\sigma(\alpha^{(j)})_k] = 2n\mu$.

Variance of $\sigma(\alpha^{(j)})_k$: Decomposing the canonical embedding double sum into diagonal ($\ell = m$) and off-diagonal ($\ell \neq m$) terms, diagonal contributions have variance $4\sigma_{\text{sign}}^2 - \mu^2$ while off-diagonal terms (independent) have variance $4\sigma_{\text{sign}}^2$. Therefore,

$$\begin{aligned} \text{Var}(\sigma(\alpha^{(j)})_k) &= 2 \left[n(4\sigma_{\text{sign}}^2 - \mu^2) + n(n-1) \cdot 4\sigma_{\text{sign}}^2 \right] \\ &= 8n^2\sigma_{\text{sign}}^2 - 2n\mu^2 \approx 8n^2\sigma_{\text{sign}}^2. \end{aligned}$$

where the $2n\mu^2$ term is negligible compared to $8n^2\sigma_{\text{sign}}^2$ for large n .

Distribution of $\beta^{(j)}$: By definition, $\mathbb{E}[\sigma(\beta^{(j)})_k] = 0$ and the same decomposition gives $\text{Var}(\sigma(\beta^{(j)})_k) = 8n^2\sigma_{\text{sign}}^2$. By CLT, both are approximately complex Gaussian. $\square$

Random vector ℓ_∞ concentration

Let $\psi \in \mathbb{R}^k$ be a random vector with $\|\psi\|_2 = 1$ whose entries are determined by k independent random samples with iid Gaussian-like behavior. Then with probability $1 - O(k^{-c})$ for any constant $c > 0$,

$$\|\psi\|_\infty \sim \sqrt{\frac{2 \log k}{k}}.$$

where $\sim$ denotes that the ratio is bounded by absolute constants.

Proof. For a normalized random vector with independent, identically distributed Gaussian-like entries, each coordinate has variance $\approx 1/k$. The maximum of k independent Gaussians with variance $\sigma^2 = 1/k$ satisfies $\mathbb{E}[\max_i |\psi_i|] \approx \sigma \sqrt{2 \log k} = \sqrt{2 \log k / k}$. With concentration, $\|\psi\|_\infty = \Theta(\sqrt{\log k / k})$ with probability $1 - O(k^{-c})$. $\square$

Lemma 4.6

Let $G_0 \in \mathbb{R}^{k \times k}$ have entries

$$(G_0)_{ij} := \text{T}((\ell^{(i)})^* Q^{-1} \ell^{(j)}),$$

where $\ell^{(j)}$ and $\mathbf{z}^{(j)}$ are as in Definition 2. Then

$$(G_0)_{ij} = \kappa^{(i)} \kappa^{(j)} \text{T}((\mathbf{z}^{(i)})^* \mathbf{z}^{(j)}).$$

In particular, under the noisy-sign model of Lemma 4.5 with flip rate $p < \frac{1}{2}$,

$$\mathbb{E}[G_0] = \sigma_0 I_k + \lambda \psi \psi^T, \quad \psi = (\kappa^{(1)}, \dots, \kappa^{(k)})^T,$$

with $\lambda = 4n^2\mu^2$ and $\sigma_0 = 16n^2\sigma_{\text{sign}}^2$, where $\mu = (1-2p)\mu_{\text{sign}}$.

Proof. From Lemma 4.5(1) and $B^* Q^{-1} B = I_2$, we obtain

$$\begin{aligned} (G_0)_{ij} &= \kappa^{(i)} \kappa^{(j)} \text{T}((\mathbf{z}^{(i)})^* \mathbf{z}^{(j)}) \\ &= \kappa^{(i)} \kappa^{(j)} (\alpha^{(i)*} \alpha^{(j)} + \beta^{(i)*} \beta^{(j)}). \end{aligned}$$

For $i \neq j$: independence gives $\mathbb{E}[\text{T}(\alpha^{(i)*} \alpha^{(j)})] = 4n^2\mu^2$ and $\mathbb{E}[\text{T}(\beta^{(i)*} \beta^{(j)})] = 0$, so $\mathbb{E}[(G_0)_{ij}] = \kappa^{(i)} \kappa^{(j)} \cdot 4n^2\mu^2$.

For $i = j$: using second moments $\mathbb{E}[|\sigma(\alpha^{(i)})_k|^2] = 4n^2\mu^2 + 8n^2\sigma_{\text{sign}}^2$ and $\mathbb{E}[|\sigma(\beta^{(i)})_k|^2] = 8n^2\sigma_{\text{sign}}^2$ gives $\mathbb{E}[(G_0)_{ii}] = 4n^2\mu^2 + 16n^2\sigma_{\text{sign}}^2$.

Letting $\psi = (\kappa^{(1)}, \dots, \kappa^{(k)})^T$, $\lambda = 4n^2\mu^2$, and $\sigma_0 = 16n^2\sigma_{\text{sign}}^2$ gives

$$\mathbb{E}[G_0] = \sigma_0 I_k + \lambda \psi \psi^T.$$

$\square$

Theorem 4.7

Given k pairs $\{(\mathbf{w}^{(j)}, I^{(j)})\}_{j=1}^k$ whose signs follow the noisy-sign model, if

$$k \geq \frac{8\sigma_{\text{sign}}^2 (\log k)^2}{n\mu^2} \left(1 + \frac{2\sigma_{\text{sign}}^2}{\mu^2} \right),$$

where $\mu = (1-2p)\mu_{\text{sign}}$, then the algorithm runs in polynomial time $T_1 = O(Tkn)$ and for each $j = 1, \dots, k$, with probability $1 - O(1/k)$, the output of Algorithm 1 satisfies

$$\text{sign}(\hat{\psi}_j) = \kappa^{(j)}.$$

Proof of Theorem 4.7. By Lemma 4.6, $\mathbb{E}[G_0] = \sigma_0 I_k + \lambda \psi \psi^T$ with eigengap $\Delta = k\lambda$. Let $E = G_0 - \mathbb{E}[G_0]$ and $\hat{\psi}$ be the top unit eigenvector. The Davis–Kahan theorem gives

$$\|\hat{\psi} - \psi / \sqrt{k}\|_2 \leq \frac{\|E\|_{\text{op}}}{\Delta} = \frac{\|E\|_{\text{op}}}{k\lambda}.$$

For off-diagonal entries $i \neq j$, computing variances using $\mathbb{E}[\sigma(\alpha^{(i)})_k]^2 = 4n^2\mu^2 + 8n^2\sigma_{\text{sign}}^2$ and $\mathbb{E}[\sigma(\beta^{(i)})_k]^2 = 8n^2\sigma_{\text{sign}}^2$, then averaging over n slots gives

$$\text{Var}((G_0)_{ij}) = 64n^3\sigma_{\text{sign}}^2(\mu^2 + 2\sigma_{\text{sign}}^2), \quad \text{for } i \neq j.$$

By the matrix Bernstein inequality with variance $\sigma^2 = 64n^3\sigma_{\text{sign}}^2(\mu^2 + 2\sigma_{\text{sign}}^2)$, $\|E\|_{\text{op}} \lesssim \sqrt{k\sigma^2 \log k}$.

For correct sign recovery, we need $\text{sign}(\hat{\psi}_j) = \text{sign}(\psi_j/\sqrt{k})$ for all j , which requires

$$|\hat{\psi}_j - \psi_j/\sqrt{k}| < |\psi_j/\sqrt{k}| = \frac{1}{\sqrt{k}}, \quad \forall j.$$

that is, $\|\hat{\psi} - \psi/\sqrt{k}\|_{\infty} < 1/\sqrt{k}$.

By the ℓ_{∞} concentration lemma,

$$\|\hat{\psi} - \psi/\sqrt{k}\|_{\infty} \lesssim \|\hat{\psi} - \psi/\sqrt{k}\|_2 \cdot \sqrt{\frac{2 \log k}{k}}.$$

Combining with Davis-Kahan gives $\|E\|_{\text{op}} < \Delta/\sqrt{2 \log k}$. Using matrix Bernstein $\|E\|_{\text{op}} \lesssim \sqrt{k\sigma^2 \log k}$,

$$\sqrt{k\sigma^2 \log k} < \frac{4kn^2\mu^2}{\sqrt{2 \log k}}.$$

Squaring and simplifying yields $2k\sigma^2(\log k)^2 < 16k^2n^4\mu^4$. Substituting $\sigma^2 = 64n^3\sigma_{\text{sign}}^2(\mu^2 + 2\sigma_{\text{sign}}^2)$ and $\Delta = 4kn^2\mu^2$ gives

$$k > \frac{8(\log k)^2\sigma_{\text{sign}}^2}{n\mu^2} \left(1 + \frac{2\sigma_{\text{sign}}^2}{\mu^2}\right).$$

For each entry j , the concentration bounds give a failure probability of $O(1/k)$.

For the time complexity: the algorithm uses the Lanczos method to compute the top eigenvector of G_0 without forming G_0 explicitly. Instead, it performs matrix-vector products using $G_0 = A \text{Lift}(Q^{-1})A^{\top}$, where each multiplication takes $O(kn)$ time. With T Lanczos iterations, the total complexity is $T_1 = O(Tkn)$. $\square$

Theorem 4.8

Let B be the secret key and $Q = B^*B$ the public key of HAWK. Given k pairs $\{(\hat{\mathbf{w}}^{(j)}, I^{(j)})\}_{j=1}^k$ with exact sign observations ($p = 0$), the success probability of recovering B in time $T_2 = O(n^3k)$ is determined by whether the observed hyperplanes are in general position. Specifically,

$$\Pr[\text{key recovery} \mid k] = \mathbb{E}_{m \sim \text{Bin}(2nk, p_0)}[P_{2n-1}(m)]$$

where $p_0 = \frac{1}{2} \Pr_{Z \sim D_{Z, \sigma_{\text{sign}}}}[Z = 0]$ and

$$P_d(m) := 1 - 2^{-m+1} \sum_{j=0}^{d-1} \binom{m-1}{j}$$

is the probability that m random hyperplanes through the origin in $\mathbb{R}^d$ are in general position (Wendel [22]).

Proof. Let $\mathbf{z}_{\text{key}} := \text{Vec}(G, -g) \in \mathbb{Z}^{2n}$ denote the first column of B^{-1} . From sign observations, construct constraint matrices $A^{(j)} := \text{Diag}(\mathbf{v}^{(j)}) \cdot \text{Lift}(M_{\mathbf{w}}^{(j)})$ where $\mathbf{v}^{(j)}$ encodes the observed signs. Stacking these forms $A \in \mathbb{R}^{2nk \times 2n}$ with feasible cone $\mathcal{K} := \{\mathbf{z} \in \mathbb{R}^{2n} : A\mathbf{z} \geq 0\}$.

Let m be the number of zero coefficients. The cone $\mathcal{K}$ collapses to a ray if and only if the m zero-coefficient constraints are in general position.

By Wendel's theorem [22], m random hyperplanes in $\mathbb{R}^d$ partition into 2^m regions with probability $P_d(m) = 1 - 2^{-m+1} \sum_{j=0}^{d-1} \binom{m-1}{j}$. Since $m \sim \text{Bin}(2nk, p_0)$, the success probability is

$$\Pr[\text{key recovery} \mid k] = \mathbb{E}_{m \sim \text{Bin}(2nk, p_0)}[P_{2n-1}(m)].$$

For time complexity: constructing $A \in \mathbb{R}^{2nk \times 2n}$ takes $O(n^2k)$ time, and solving via Algorithm 4 takes $O(n^3k)$. The overall complexity is $T_2 = O(n^3k)$. $\square$

Lemma 4.9

Let $\ell^{(j)}$ and $\mathbf{z}^{(j)}$ be the observable and theoretical feature vectors. The aligned aggregates

$$\ell_{\Sigma} := \sum_{j=1}^k \mathcal{X}^{(j)} \ell^{(j)}, \quad \mathbf{z}_{\Sigma} := \sum_{j=1}^k \mathbf{z}^{(j)}.$$

satisfy the algebraic relation $\ell_{\Sigma} = B^* \mathbf{z}_{\Sigma}$.

Proof. Follows by summing the relations $\ell^{(j)} = \mathcal{X}^{(j)} B^* \mathbf{z}^{(j)}$ from Lemma 4.5 and using $(\mathcal{X}^{(j)})^2 = 1$. $\square$

Theorem 4.10

Suppose the symmetry-breaking signs $\{\mathcal{X}^{(j)}\}_{j=1}^k$ have been correctly recovered. If

$$k \geq \frac{12C\eta\sigma_{\text{sign}}^2 \log n}{\mu^2},$$

then the aggregation and slot-wise normalization procedure recovers f^* (and hence the secret basis B) in time $T_3 = O(kn + n \log n)$ via coefficient-wise rounding, with probability at least $1 - n^{1-\Omega(C)}$, where $C > 0$ is an absolute constant controlling the confidence level.

Proof. The algorithm is given in Algorithm 3. We analyze why it succeeds and derive the sample complexity bound.

Key algebraic relation. By Lemma 4.9, $\ell_{\Sigma} = B^* \mathbf{z}_{\Sigma}$ where $\mathbf{z}_{\Sigma} = (\alpha_{\Sigma}, \beta_{\Sigma})$. At slot s , normalized feature $(u_s^{(1)}, u_s^{(2)})$ gives first component $f^*(\zeta^s) \cdot u_s^{(1)} + g^*(\zeta^s) \cdot u_s^{(2)}$.

High-SNR Taylor expansion and variance analysis. Write $\sigma(\alpha_{\Sigma})_s = \mu_{\alpha} + Z_{\alpha}$ and $\sigma(\beta_{\Sigma})_s = Z_{\beta}$ where $\mu_{\alpha} := 2kn\mu$ and $\mathbb{E}[|Z_{\alpha}|^2] = \mathbb{E}[|Z_{\beta}|^2] = 8kn^2\sigma_{\text{sign}}^2$. In high SNR ($|\mu_{\alpha}| \gg \sqrt{\mathbb{E}[|Z_{\alpha}|^2]}$), Taylor expansion gives

$$u_s^{(1)} \approx 1 + \delta_s^{(1)}, \quad u_s^{(2)} \approx \frac{Z_{\beta}}{\mu_{\alpha}}.$$

where $\delta_s^{(1)} := \frac{\text{Im}(Z_\alpha)}{\mu_\alpha} - \frac{|Z_\beta|^2}{2\mu_\alpha^2}$. Since Z_α, Z_β are independent complex Gaussians,

$$\mathbb{E}[|\text{Im}(Z_\alpha)|^2] = \frac{1}{2} \mathbb{E}[|Z_\alpha|^2] = 4kn^2\sigma_{\text{sign}}^2.$$

hence the first-order error has variance

$$\mathbb{E}[\delta_s^{(1)^2}] \approx \frac{\mathbb{E}[|\text{Im}(Z_\alpha)|^2]}{\mu_\alpha^2} = \frac{4kn^2\sigma_{\text{sign}}^2}{(2kn\mu)^2} = \frac{\sigma_{\text{sign}}^2}{k\mu^2}.$$

The second component has variance

$$\mathbb{E}[|u_s^{(2)}|^2] = \frac{\mathbb{E}[|Z_\beta|^2]}{\mu_\alpha^2} = \frac{8kn^2\sigma_{\text{sign}}^2}{(2kn\mu)^2} = \frac{2\sigma_{\text{sign}}^2}{k\mu^2}.$$

Error decomposition. The first component of $B^*(\zeta^s) \begin{pmatrix} u_s^{(1)} \\ u_s^{(2)} \end{pmatrix}$ yields $f^*(\zeta^s) \cdot u_s^{(1)} + g^*(\zeta^s) \cdot u_s^{(2)} = f^*(\zeta^s) + \varepsilon_s$ where $\varepsilon_s := f^*(\zeta^s)\delta_s^{(1)} + g^*(\zeta^s)u_s^{(2)}$. From the variance results above,

$$\text{Var}(\varepsilon_s) = |f^*(\zeta^s)|^2 \cdot \frac{\sigma_{\text{sign}}^2}{k\mu^2} + |g^*(\zeta^s)|^2 \cdot \frac{2\sigma_{\text{sign}}^2}{k\mu^2}.$$

Inverse embedding and ℓ_2 norm via Parseval. By Parseval, $\|\hat{\varepsilon}\|_2^2 = \frac{1}{n} \sum_s |\varepsilon_s|^2$, so taking expectations and using Parseval's identity $\sum_s |f^*(\zeta^s)|^2 = n\|f^*\|_2^2$ gives

$$\mathbb{E}[\|\hat{\varepsilon}\|_2^2] = \frac{\sigma_{\text{sign}}^2}{k\mu^2} (\|f^*\|_2^2 + 2\|g^*\|_2^2) \approx \frac{3n\eta\sigma_{\text{sign}}^2}{2k\mu^2}.$$

using $\|f^*\|_2^2 \approx \|g^*\|_2^2 \approx n\eta/2$.

ℓ_∞ bound and rounding condition. By the ℓ_2 -to- ℓ_∞ concentration bound,

$$\|\hat{\varepsilon}\|_\infty \leq \sqrt{\frac{2C \log n}{n}} \|\hat{\varepsilon}\|_2 = \sqrt{\frac{3C\eta\sigma_{\text{sign}}^2 \log n}{k\mu^2}}.$$

For coefficient-wise rounding to succeed, we require $\|\hat{\varepsilon}\|_\infty < 1/2$:

$$\sqrt{\frac{3C\eta\sigma_{\text{sign}}^2 \log n}{k\mu^2}} < \frac{1}{2}.$$

Squaring and solving for k gives

$$k > \frac{12C\eta\sigma_{\text{sign}}^2 \log n}{\mu^2}.$$

Once f is recovered, the remaining key components follow from algebraic relations.

Time complexity. Given precomputed features, aggregation takes $O(kn)$ and FFT-based operations take $O(n \log n)$, yielding $T_3 = O(kn + n \log n)$. $\square$

Appendix B. Interior point method

This appendix describes the primal-dual interior point method (IPM) used to solve the linear program in Category II (Section 4.4).

B.1. Linear Programming Formulation

Using the constraint matrix $A \in \mathbb{R}^{m \times 2n}$ (where $m = 2nk$) from Definition 3, we seek $\mathbf{z} \in \mathbb{R}^{2n}$ satisfying $A\mathbf{z} \succeq \mathbf{0}$. Since these constraints define a cone through the origin, we introduce normalization constraint $\langle \mathbf{c}_{\text{sum}}, \mathbf{z} \rangle = 1$ where $\mathbf{c}_{\text{sum}} := \sum_{i=1}^m \mathbf{a}_i$. The LP formulation is:

$$\max_{\mathbf{z}} \langle \mathbf{c}, \mathbf{z} \rangle \quad \text{s.t.} \quad A\mathbf{z} \succeq \mathbf{0}, \langle \mathbf{c}_{\text{sum}}, \mathbf{z} \rangle = 1, \quad (8)$$

where $\mathbf{c} \in \mathbb{R}^{2n}$ is orthogonal to $\mathbf{c}_{\text{sum}}$. This recovers the direction of $\mathbf{z}_{\text{key}}$ on the cone boundary, from which B is reconstructed.

B.2. KKT Conditions and Barrier Method

The Karush-Kuhn-Tucker (KKT) conditions characterize the optimality of the LP. Introducing slack variables $\mathbf{s} \in \mathbb{R}^m$, dual variables $\mathbf{z}_d \in \mathbb{R}^m$, and Lagrange multiplier $\nu \in \mathbb{R}$, the KKT system is:

$$\begin{aligned} A\mathbf{z} - \mathbf{s} &= \mathbf{0}, & -A^\top \mathbf{z}_d + \mathbf{c}_{\text{sum}}\nu &= \mathbf{c}, & \langle \mathbf{c}_{\text{sum}}, \mathbf{z} \rangle &= 1, \\ s_i(z_d)_i &= 0 \quad \forall i, & \mathbf{s}, \mathbf{z}_d &\succ \mathbf{0}. \end{aligned} \quad (9)$$

The barrier method solves this system iteratively by relaxing the complementarity condition $s_i(z_d)_i = 0$ to $s_i(z_d)_i \approx \sigma\mu$ for all i , where $\mu := \langle \mathbf{s}, \mathbf{z}_d \rangle / m$ is the duality gap and $\sigma \in (0, 1)$ is a centering parameter (typically $\sigma = 0.4$). As $\mu \rightarrow 0$ through successive iterations, exact complementarity is recovered.

B.3. Newton's Method for the KKT System

Applying Newton's method and eliminating $\Delta \mathbf{s}$ and $\Delta \mathbf{z}_d$ reduces to the $(2n+1) \times (2n+1)$ system:

$$\begin{bmatrix} A^\top W A & \mathbf{c}_{\text{sum}} \\ \mathbf{c}_{\text{sum}}^\top & 0 \end{bmatrix} \begin{bmatrix} \Delta \mathbf{z} \\ \Delta \nu \end{bmatrix} = \begin{bmatrix} -\mathbf{b} \\ -r_e \end{bmatrix}, \quad (10)$$

where $W := Z_d S^{-1}$ is diagonal ($S = \text{diag}(\mathbf{s})$, $Z_d = \text{diag}(\mathbf{z}_d)$), and $\mathbf{b}, r_e$ combine KKT residuals. After solving, $\Delta \mathbf{s}$ and $\Delta \mathbf{z}_d$ are recovered.

B.4. Algorithm

Algorithm 4 presents the IPM. Exploiting diagonal W reduces per-iteration complexity from $O(n^3 k^3)$ to $O(n^3 k)$. Typically converges in 20–50 iterations.

Algorithm 4: Primal-Dual IPM for Category II LP

Input: Constraint matrix $A \in \mathbb{R}^{m \times 2n}$

Output: Feasible $z^* \in \mathbb{R}^{2n}$ with $Az^* \succeq 0$,
 $\langle c_{\text{sum}}, z^* \rangle = 1$

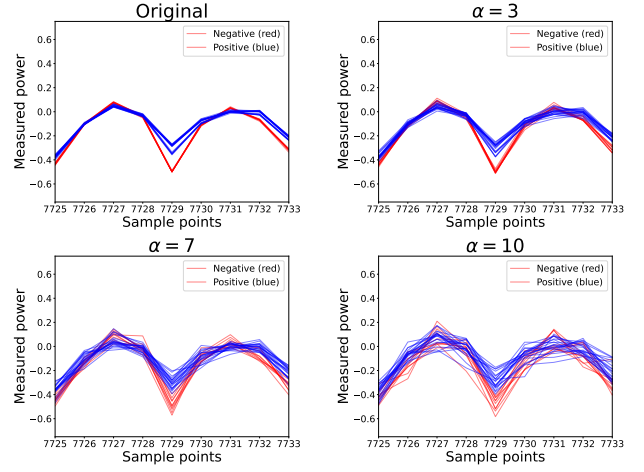
- 1 Compute $c_{\text{sum}} \leftarrow \sum_{i=1}^m a_i$ (sum of rows of A)
 - 2 Choose $c \in \mathbb{R}^{2n}$ orthogonal to c_{sum} (e.g., random vector in $c_{\text{sum}}^\perp$)
 - 3 Initialize z, s, z_d, ν with $s, z_d \succ 0$; set $\sigma = 0.4$, $\tau = 0.99$
 - 4 **for** $\text{iter} = 1, \dots, \text{max_iter}$ **do**
 - 5 Compute residuals r_p, r_d, r_e and duality gap
 $\mu = \langle s, z_d \rangle / m$
 - 6 **if** $\min(Az) \geq -\varepsilon(1 + \|z\|)$ **then**
 - 7 **return** z
 - 8 Compute $W = Z_d S^{-1}$, $r_\mu = S Z_d - \sigma \mu \mathbf{1}$,
 $b = r_d + A^\top (S^{-1} r_\mu + W r_p)$
 - 9 Solve (10) for $(\Delta z, \Delta \nu)$
 - 10 $\Delta s \leftarrow A \Delta z + r_p$;
 $\Delta z_d \leftarrow (-r_\mu - Z_d \Delta s) \oslash s$
 - 11 Compute step sizes α_p, α_d via
fraction-to-boundary rule
 - 12 Update: $z \leftarrow z + \alpha_p \Delta z$, $s \leftarrow s + \alpha_p \Delta s$,
 $z_d \leftarrow z_d + \alpha_d \Delta z_d$, $\nu \leftarrow \nu + \alpha_p \Delta \nu$
 - 13 **return** z
-

Appendix C.

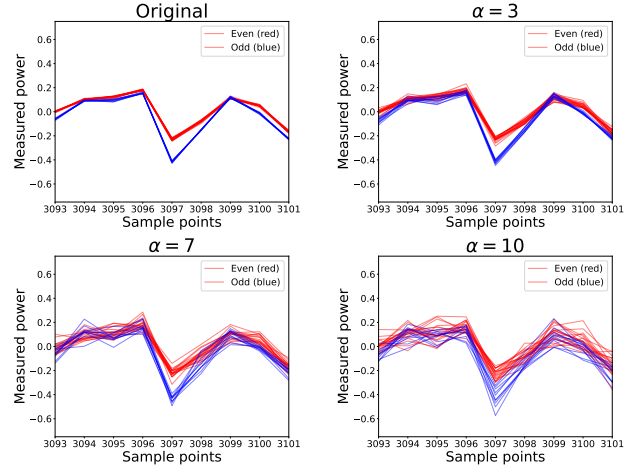
Empirical Visualization of HAWK's DGS Leakage under Noise

Figure 1 visualizes how side-channel leakages manifest and degrade across different noise conditions. For each leakage type, we plot the original trace ($\alpha = 0$) alongside three noisy cases ($\alpha = 3, 7, 10$), where each panel is generated from 30 traces collected under the corresponding noise setting. The color of each trace encodes the underlying coefficient information. For instance, negative coefficients are shown in red and positive ones in blue, which allows visual distinction of how the corresponding data property influences the power consumption. The horizontal axis corresponds to the sample index and can be interpreted as a time axis, while the vertical axis represents the measured power consumption. Each panel zooms in around the sampling instant at which the corresponding information is leaked.

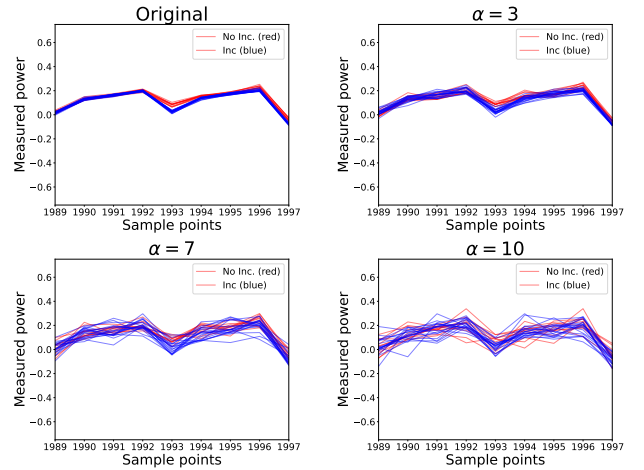
For *sign* leakage, the difference between negative and positive coefficients becomes visible around sample point 7729. For *parity* leakage, the gap between even and odd coefficients emerges around sample point 3097. For *increment* leakage, the effect of the table comparison, namely whether it causes the coefficient value to increase, can be observed around sample point 1993. This arrangement allows direct visual comparison of how the information content at these leakage points progressively diminishes as the noise level increases.



(a) Sign leakage in HAWK's DGS



(b) Parity leakage in HAWK's DGS



(c) Increment leakage in HAWK's DGS

Figure 1. Visualization of sign, parity, and increment leakages in HAWK's DGS under the original trace ($\alpha = 0$) and three noisy levels ($\alpha = 3, 7, 10$), with each subfigure showing how leakage around the marked sample points fades as noise increases.